

**TECHNICAL SPECIFICATION
FOR BUILDING FACILITY
MANAGEMENT & MAINTENANCE**

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	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	TABLE OF CONTENTS	Section Revision: 2
		Page 1 of 85

DEFINITION	4
SECTION A: GENERAL.....	9
1.0 PURPOSE	9
2.0 OBJECTIVE.....	9
3.0 SCOPE OF SERVICES	10
4.0 SITE INFORMATION AND ASSET INVENTORIES.....	12
4.1 As-Built Drawing.....	12
4.2 Asset & Spares Inventories	13
4.3 Operation & Maintenance (O&M) Manuals	14
4.4 Testing and Commissioning Record	14
5.0 RULES, REGULATIONS AND STANDARDS	14
SECTION B: FACILITY MANAGEMENT PLAN (FMP)	17
6.0 FACILITY MANAGEMENT PLAN (FMP)	17
SECTION C: CONTRACTOR SITE ORGANISATION.....	20
7.0 FMM WORK PROGRAM.....	20
8.0 MONTHLY PAYMENT PLAN (MPP).....	22
9.0 HUMAN RESOURCE REQUIREMENT	23
9.1 Staffing / Staff Requirement.....	23
9.2 Competency of Operation and Maintenance Staff	24
9.3 Working Hours.....	27
9.4 Safety, Health and Environment (SHE) of the Staff	28
10.0 QUALITY MANAGEMENT SYSTEM	29
10.1 Quality Management System.....	29
11.0 MATERIALS, SPARES, TOOLS & EQUIPMENT REQUIREMENT	30
12.0 CONTRACTOR SITE OFFICE, INSPECTION TOOLS, EQUIPMENT & FACILITIES	31
SECTION D: FACILITY MANAGEMENT SERVICES.....	32
13.0 TRANSITION MANAGEMENT	32
13.1 Transition-In Management Plan.....	32
13.2 Transition-Out Management Plan.....	33
13.3 Facility Familiarization and Verification (FFV).....	34
13.4 Facility Condition Assessment (FCA)	34
13.5 Handover of the Asset.....	35

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	TABLE OF CONTENTS	Section Revision: 2
		Page 2 of 85

14.0 CUSTOMER CARE MANAGEMENT	36
14.1 Customer Care Service	36
14.2 Help Desk Personnel.....	39
14.3 Liaison / Communication	40
15.0 MANAGEMENT INFORMATION SYSTEM (MIS)	40
15.1 Management Information System (MIS) Plan.	40
15.2 Application Management	41
15.3 Technical Library & Repository	41
15.4 Repository	42
15.5 Warranty Management	42
16.0 UTILITIES MANAGEMENT	43
16.1 Electricity Supply	43
16.2 Cold Water Supply	44
16.3 Chilled Water Supply (if applicable)	45
16.4 Treated Sewerage Effluent (TSE).....	45
16.5 Sewerage Collection and Disposal Service	46
16.6 LPG / CNG Supply	46
17.0 FACILITY RISK MANAGEMENT	47
18.0 FACILITIES HEALTH, SAFETY AND ENVIRONMENTAL (HSE) MANAGEMENT ..	48
19.0 INCIDENT RESPONSE AND DISASTER RECOVERY MANAGEMENT (IRDRM) ..	52
20.0 ENERGY MANAGEMENT	54
SECTION E: FACILITY ENGINEERING OPERATION AND MAINTENANCE SERVICES	58
21.0 GENERAL	58
22.0 OPERATION SERVICES.....	61
22.1 Help Desk Services	61
22.2 CMMS Services.....	62
22.3 Landscaping	64
22.4 Waste and Redundant Materials	65
23.0 MAINTENANCE SERVICES.....	66
23.1 Maintenance Schedule and Frequency	66
23.2 Verification of Completed Work / Deduction	66
23.3 Parts and Consumables Item	66
23.4 Mechanical Services	67
23.5 Electrical Services	71

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	TABLE OF CONTENTS	Section Revision: 2
		Page 3 of 85

23.6 Civil, Structural & Architectural Services.....	75
SECTION F: FACILITY CUSTODIAL SERVICES	79
24.0 HOUSEKEEPING SERVICES	79
24.1 Cleaning Equipment	80
24.2 Cleaning Machine.....	80
24.3 Chemical Cleaning	81
24.4 Hygienic Services.....	81
25.0 PEST CONTROL SERVICES.....	82
SECTION G: TECHNICAL ADVICE AND EXPERT PROPOSAL SERVICES	85
26.0 TECHNICAL ADVICE AND EXPERT PROPOSAL SERVICES	85

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	DEFINITION	Section Revision: 2
		Page 4 of 85

DEFINITION

Asset	All assets to enable the Contractor to provide FMM Services in accordance with this technical specification, including but not limited to: - 1. any land or buildings and infrastructure; 2. plant and machineries; 3. equipment and system (together with any warranties in respect of assets being transferred); 4. drawings and technical data; 5. intellectual property rights as attached to assets; 6. any books, reports and records which including all operating and maintenance manuals; and 7. any spare parts, tool and other assets (together with any warranties in respect of assets being transferred).
Building Information Modelling (BIM)	Building Information Modelling (BIM) is a process of preparing, utilizing, managing, and coordinating project or asset information and documentation through an intelligent digital model throughout the lifecycle of a project, where this digital model is used by various parties for specific purposes to enhance the overall effectiveness of asset management.
BIM Execution Plan Asset and Facility Management (BEPAFM)	A BEPAFM is a strategic implementation plan that defines how Building Information Modelling (BIM) will be applied and managed throughout the asset and facility management phase. It serves as a guideline to ensure that all stakeholders can effectively use, share, and maintain BIM data to optimize asset performance, operational efficiency, and decision-making across the facility lifecycle.
Client	Asset owner or user / customer's representative who can make decision regarding the Asset & Facility.
Complaint	Any written or verbal notice or report by the Customers / Users / FSO / FSO's representative or NCR from Government audit activity given through telephone / fax / electronic media (or any other means agreed by FSO) about not meeting agreed

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	DEFINITION	Section Revision: 2
		Page 5 of 85

	standard of service provided by the Contractor as per scope and technical specification (after verified by the Contractor).
Contractor	Refers to FMM Contractor appointed by the Government to implement the FMM Contract.
Corrective Maintenance	Work / repair / servicing / testing carried out after failure / breakdown has occurred and is intended to restore an item to a state in which it can perform to its required function as per designed.
Custodial Services	General cleaning / housekeeping, pest & domestic animal control and hygiene services.
Defect	Any damage, failure, faulty or malfunction which may be present in a product, asset or facility.
Evaluation Month	Agreed period and cut-off date between FSO and the Contractor of the FMM Services evaluation for Monthly Payment purpose.
Execution Time	Time measured from the Complaint first logged on the Contractor's Help Desk until Work Order is closed
Facility	Collection of Assets which is built, installed or established to serve an entity's needs.
Facility Condition Assessment (FCA)	The process of inspection, assessment, and measurement to identify defects, damage, and deterioration of the entire building/component and/or building system to determine follow-up actions such as maintenance/ restoration/ renovation/ upgrading/ disposal
Facility Management and Maintenance (FMM)	Managing, administrating, supervising all aspect of asset including but not limited to facility management services, facility engineering operation & maintenance services, custodial services and technical advice & expert proposal services within the asset's boundaries.
Facility Management Plan (FMP)	A document that describes tactical and operational plan encompasses understanding, analysing, planning, and acting for the Contractor to implement FMM Services throughout the contract period

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	DEFINITION	Section Revision: 2
		Page 6 of 85

Facility Risk Management	A structured approach to managing effect of uncertainty related to threat, a sequence of human activities including risk assessment, strategies development to manage it, and mitigation of risk using managerial resources.
Facility Superintending Officer (FSO)	Facility Superintending Officer and his successor in office as appointed by the Government under Clause 23 in PWD Facilities Management & Maintenance Form (2016) to act on behalf of the Government for matters stipulated in this Contract.
Incident Report	A document that record details of an unusual and significant event(s) that occurs at the asset.
Preventive Maintenance	A work / repair / servicing and testing carried out at pre-determined intervals period to prescribed criteria, intended to reduce failure, environmental damage or the performance degradation of an item to prolong the life expectancy of the equipment.
Planned Preventive Maintenance (PPM)	A planned and scheduled work to carry out the preventive maintenance. The Planned Preventive Maintenance also includes Routine Inspection and Predictive Maintenance / Condition Based Maintenance.
Premise	A physical location or site that accommodates buildings, infrastructure, and related facilities.
Quality Management System	A collection of processes focussed on consistently meeting user / customer requirement and enhancing their satisfaction compliant with the latest MS ISO 9001. It is expressed as the organisational structure, policy, procedures, processes and resources needed to implement Quality Management.
Response Time	Time measured from the date and time a Request/Complaint is logged (Work Request open) until verification of the Complaint (Work Order open)
Scheduled Corrective Maintenance (SCM)	Scheduled Corrective Maintenance (SCM) is a planned maintenance activity scheduled to rectify deficiencies or failures identified through PPM, routine inspections, or user complaints. The activities may involve repair, restoration, or replacement of equipment, and are carried out according to

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	DEFINITION	Section Revision: 2
		Page 7 of 85

	required timelines, availability of spare parts, or during routine maintenance shutdowns.
Sub-contractor	Company/specialist/service provider hired by a FMM contractor to carry out specific tasks or services within the FMM contract. The sub-contractor works under the direction of the FMM contractor, who remains responsible for the overall FMM Services
Technical Advice Report	A document/report that describes the status, process, progress, or results of technical or scientific research carried out by qualified personnel which include recommendations and conclusions to improve the FMM Services.
Transition Management	Managing all aspects of transition period which includes transition of the asset from the construction Contractor to the FMM Contractor; and from the current FMM Contractor to the new succeeding FMM Contractor.
Transition-in	The maximum period of three (3) months from Commencement Date for the successor Contractor to carry out activities involving gathering all the resources to operate and maintain the asset
Transition-out	The period of three (3) months before the expiry of the FMM Contract Period for the Contractor to carry out activities involving shedding / minimizing the resources to operate and maintain an asset, clearing all work order issued and making good all defects prior to the end date of a contract, training and handing over to the successor Contractor.
User / Customer	All staff, tenants, visitors and / or occupiers who will reside, operate and / or have their place of business or work at the asset sites. User / customer shall include but not limited to Government agencies, bodies and / or authorities.
Work order	A Work Order is a formal instruction issued once a Work Request or complaint has been verified. It specifies the details of the work to be carried out, including type of work, execution timeline, reference number, status, cost, and customer confirmation. Work Orders can also be generated from Routine

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	DEFINITION	Section Revision: 2
		Page 8 of 85

	Inspections or Self-Finding by the Contractor's staff without any user complaint.
Work request	A Work Request is an initial complaint or service request submitted through the Help Desk, either by users, customers, the FSO or the FSO's representative. It is recorded in the system (CMMS or manual), and the Response Time is measured from the moment the request/complaint is logged until it is verified and converted into a Work Order.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
		Section Revision: 2
SECTION A:	GENERAL	Page 9 of 85

SECTION A: GENERAL

1.0 PURPOSE

- 1.1 The Facility Management and Maintenance Contract (FMMC) is focusing on:
- a) sustaining asset value;
 - i. optimizing asset usage;
 - ii. optimizing operational and maintenance costs
 - b) supporting organisation service delivery; and
 - c) delighting user/customer's experience.
- 1.2 This FMMC shall be a single point of responsibility in all defined FMM Services, which the Contractor shall give full commitment, ownership, proactive manner and accountability to ensure the quality of service of the here said facility.

2.0 OBJECTIVE

- 2.1 The Contractor shall carry out Facility Management & Maintenance (FMM) Services according to the scope of services for the asset and facilities within the site boundaries to be determined by the Government to meet the objective of the FMMC as stated below:
- a) to ensure asset at optimum functionality;
 - b) to ensure asset at optimum operations;
 - c) to optimize the operational and maintenance costs of the asset;
 - d) to maintain asset service level performance;
 - e) to comply all performance requirements of FMM scope;
 - f) to fulfil the satisfactory level of user/customer;
 - g) to contribute recommendations through facility process improvement for client to increase efficiency in their core business delivery;
 - h) to ensure availability of the latest data and information in the accepted Management Information System (MIS) which shall be compatible with and able to integrate with the Government's asset management system; and to comply with the current Government's asset policies and procedures such as Dasar Pengurusan Aset Kerajaan (DPAK), Manual Pengurusan Aset Menyeluruh (MPAM) and Tatacara Pengurusan Aset Tak Alih (TPATA)

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
		Section Revision: 2
SECTION A:	GENERAL	Page 10 of 85

- i) to maintain and update the Building Information Modelling (BIM) data, including semantic and geometric information, to reflect any changes to the asset during the implementation of the FMMC, in accordance with the latest JKR guidelines and standards (if applicable).

3.0 SCOPE OF SERVICES

3.1 The following section details out the services requirements for which the Contractor must undertake to provide a satisfactory level of service delivery of the FMM Services. The Contractor shall expect to undertake, successfully implement and achieve these requirements through a process of employing best practices technique, consultation with FSO and shall be continually strived to improve the level of services.

3.2 The Contractor shall comply with all Government policies and instructions related to manage the FMMC.

3.3 The FMM Services that shall be provided, but not limited to the following:

Scope of Services	*Item
A. Facility Management Services	i. Quality Management ii. Transition Management iii. Customer Care Management iv. Management Information System (MIS) v. Utilities Management vi. Facility Risk Management vii. Human Resource Management viii. Facility Health, Safety and Environmental (HSE) Management (HSE) ix. Incident Response and Disaster Recovery Management (IRDRM) x. Energy Management xi.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
SECTION A:	GENERAL	Section Revision: 2
		Page 11 of 85

B. Facility Engineering Operation & Maintenance Services	Operation Services i. Computerised Maintenance Management System (CMMS) ii. Landscaping iii. Waste and Redundant Material iv. Maintenance Services i. Mechanical Services ii. Electrical Services iii. Civil, Structural & Architectural Services iv.
	C. Facility Custodial Services
D. Technical Advice and Expert Proposal Services	i. Housekeeping Services ii. Pest Control Services iii. i. Specialist Work ii. Procurement documentation iii. Landscape design iv. Interior design v. Building conservation vi. Building retrofitting, etc. vii.

Table 1: Scope of services

() Delete any item if not applicable*

- 3.4 The scope of services shall include management, operation and maintenance services to be delivered in relation to the asset and facilities as described in the subsequent sections below. It generally includes the following:
- the required performance from the management of the services;
 - the required service levels during the management of the services; and
 - the performance indicators that shall be used to assess the Contractor's performance.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
		Section Revision: 2
SECTION A:	GENERAL	Page 12 of 85

3.5 Management Role of the Contractor:

- a) The Contractor shall proactively manage and administer the FMM Services of the entire asset according to the scope of work within the premise as specified in the Appendix 8 and / or Appendix 10.
- b) The following utilities shall be handed over to the Contractor prior to operation commencement. The Contractor shall manage and administer the operation, maintenance and metering of utilities on behalf of the FSO for the following services:
 - i. electricity supply and distribution;
 - ii. cold water supply and distribution;
 - iii. chilled water supply and distribution (secondary circuit metering only);
 - iv. sewage collection and disposal;
 - v. Treated Sewage Effluent (TSE) / Sewage Treatment Plant (STP);
 - vi. Liquefied Petroleum Gas (LPG) / Natural Gas (CNG) supply and distribution; and
 - vii. Other utility services.
- c) The Contractor shall liaise with all utility providers from time to time.
- d) The Contractor shall manage all enquiries relating to the services.

4.0 SITE INFORMATION AND ASSET INVENTORIES

The Contractor may request a set of all relevant drawings, asset & spares inventories, operation and maintenance manuals and testing & commissioning records that are in its possession to enable the Contractor to make copies of the same for their use. Copies made shall remain the property of the Government and shall be surrendered at the end of the contract or when the contract is terminated.

4.1 As-Built Drawing

4.1.1 As-built drawing consist of the following but not limited to:-

- a) Architecture drawing
- b) Civil & Structure drawing
- c) Mechanical drawing
- d) Electrical drawing

4.1.2 Where as-built model (BIM) is provided under this Contract, all as-built drawings shall be generated directly from the updated as-built model

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
		Section Revision: 2
SECTION A:	GENERAL	Page 13 of 85

(BIM). The Contractor shall update and maintain the as-built model (BIM) in accordance with the latest JKR BIM Guidelines to accurately reflect the as-built condition of the facilities. Where BIM is not applicable under this Contract, the Contractor shall update and maintain conventional 2D as-built drawings to record all changes or modifications made during the contract period.

- 4.1.3 All the coding in as-built drawing and as-built BIM Model (if applicable) shall comply with the Sistem Kod Aset Tak Alih (SKATA) coding system requirements if applicable.
- 4.1.4 All as-built drawings/as-built model (BIM) shall be prepared using compatible software (latest release or earlier version accepted by the FSO) and a set of the complete drawings on digital format / soft copy shall also be provided.
- 4.1.5 All such drawings and model shall be stored in the Management Information System (MIS) and update whenever there are changes to the services.
- 4.1.6 Electrical single line drawing frame shall be displayed in Electrical switch rooms and being updated if there are any changes.
- 4.1.7 Mechanical single line drawing frame shall be displayed in Mechanical services room and being updated if there are any changes.
- 4.1.8 The Contractor shall supply one (1) set of softcopy and two (2) sets of hardcopy the up-to date as-built drawing according to Garis Panduan Pengurusan Lukisan Siap Bina.

4.2 Asset & Spares Inventories

- 4.2.1 The Contractor shall manage, maintain and update asset inventories and related data including spare parts inventories and utilize all Government issued spare parts (if any) in accepted Management Information System (MIS).
- 4.2.2 The Contractor shall provide the access to any of the asset inventories records that relate to the services at any time required by the FSO.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
		Section Revision: 2
SECTION A:	GENERAL	Page 14 of 85

4.3 Operation & Maintenance (O&M) Manuals

4.3.1 Contractor may obtain all related Operation & Maintenance (O&M) manuals from FSO / FSO Representative before operation commence.

4.3.2 The Contractor shall manage and update the O&M manuals if necessary.

4.4 Testing and Commissioning Record

4.4.1. All testing and commissioning results shall be documented in both softcopy and hardcopy formats and shall be submitted to FSO.

4.4.2 The test result data shall be updated into accepted MIS.

5.0 RULES, REGULATIONS AND STANDARDS

5.1 All work performed including personnel materials and equipment shall comply with the requirements, rules and regulations of but not limited to the following:

- a) Suruhanjaya Tenaga (ST),
- b) Tenaga Nasional Berhad (TNB),
- c) Suruhanjaya Komunikasi dan Multimedia (SKMM),
- d) Telekom Malaysia Berhad (TM),
- e) Jabatan Keselamatan dan Kesihatan Pekerjaan (JKKP),
- f) Suruhanjaya Perkhidmatan Air Negara (SPAN),
- g) Pihak Berkuasa Air Negeri,
- h) Jabatan Bomba dan Penyelamat,
- i) Jabatan Alam Sekitar (JAS),
- j) Jabatan Kerja Raya (JKR),
- k) Jabatan Penerbangan Awam (DCA),
- l) Lembaga Perlesenan Tenaga Atom (AELB),
- m) Lembaga Racun MakhluK Perosak,
- n) Pihak Berkuasa atau Kerajaan Tempatan,
- o) Jabatan Lanskap Negara,
- p) Pusat Racun Negara,
- q) Jabatan Kimia Malaysia,
- r) Standard and Industrial Research Institute of Malaysia (SIRIM),
- s) Ketua Pegawai Keselamatan Kerajaan (CGSO) and

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
		Section Revision: 2
SECTION A:	GENERAL	Page 15 of 85

- t) Other Government Departments, utility authorities / local authorities, Licensees and Companies having jurisdiction over the services and facilities.

5.2 In addition, the following regulations / standards shall be complied with any current amendments of acts / regulations / standards by Government, utility authorities / local authorities, Licensees and Companies having jurisdiction over the services and facilities in accordance with and subject to FSO's approval but not limited to:

- a) The Electricity Supply Act 1990 (Act 447),
- b) The Electricity Regulations 1994,
- c) MS IEC 60364 or BS 7671,
- d) Uniform Building By-Laws 1984,
- e) Occupational Safety and Health 1994 (Act 514),
- f) Fire Services Act 1998 (Act 341),
- g) Strata Management Act 2013 (Act 757),
- h) Akta Racun Makhluk Perosak 1974 (Act 149)
- i) JKR Technical Specifications
- j) Jawatankuasa Pembangunan Standard Industri JKR
- k) Perintah Gaji Minimum
- l) The related Malaysian Standards and/or IEC Standards (latest version),
- m) Other related Government Regulations

5.3 The Contractor shall be responsible for the submission of all the necessary applications to the relevant utility authorities, local authorities, other relevant authorities and licensees in connection with the execution of the works in this contract on behalf of the Government.

5.4 Any penalty or surcharge imposed by the utility authorities / licensees / companies and / or local authorities shall be borne by the Contractor due to:

- a) The Contractor's failure to comply with utility authorities / companies and / or local authorities requirements and / or,
- b) low power factor and / or non-compliance with power quality requirements due to the Contractor's negligence.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
		Section Revision: 2
SECTION A:	GENERAL	Page 16 of 85

- 5.5 All spares or replacement parts shall be new, unused and of the same make as the parts to be replaced. In the event where an alternative replacement part must be used, it shall have similar or better characteristics / features than the original part and be subject to the approval of the FSO.
- 5.6 The whole of the buildings and the services shall at all-time be in a safe and functional condition, free from danger to life and property. All work on the services shall be performed by competent personnel in compliance with the relevant laws and regulations.
- 5.7 Potential hazard related to fire shall at all-time be monitored and immediate corrective action needs to be taken. In this regard, special attention shall be given to switch boards, distribution boards, control boards, switchgear and cable / wiring terminations and fittings.
- 5.8 The Contractor shall submit monthly report for any statutory compliance requirements related to the machineries, equipment and personnel.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
SECTION B:	FACILITY MANAGEMENT PLAN (FMP)	Section Revision: 2
		Page 17 of 85

SECTION B: FACILITY MANAGEMENT PLAN (FMP)

6.0 FACILITY MANAGEMENT PLAN (FMP)

- 6.1 The Contractor shall develop, execute, implement, and maintain appropriate FMP for every component of the services.
- 6.2 Each FMP shall be a quality assured document prepared in accordance with the latest ISO. The FMP shall be treated as a live document, where any improvements are to be reviewed during the monthly meeting with the FSO to ensure continuous effectiveness and relevance. In addition, the Contractor shall ensure that the FMP is revised annually in line with the latest requirements.
- 6.3 The Contractor shall develop and submit each FMP referred to in the Scope of Services for review, comment and acceptance by the FSO. The FMP shall be consistent and integrated with all other FMP developed by the Contractor and the Government's policies and plans.
- 6.4 The Contractor shall upon each revision submit the revised FMP to the FSO, in one (1) softcopy and three (3) hardcopies, unless otherwise specified in the contract or as required by the FSO.
- 6.5 The FMP is document that describes tactical and operational plan for the Contractor to implement FMM Services. FMP development is an on-going process which encompasses understanding, analysing, planning and acting throughout the contract period. It shall incorporate an understanding of the organisation's culture and core values. FMP shall identify vital information (e.g. the type, quantity, location of spaces etc.) that are required to fully support the organisations core business initiatives and should be framed within the organisation's vision. It also includes analysis on how the facilities can manifest that culture and core values within the physical space or support service through an in-depth analysis of the facilities including location, capability, utilization and condition. The FMP shall also include an achievable and

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
		Section Revision: 2
SECTION B:	FACILITY MANAGEMENT PLAN (FMP)	Page 18 of 85

workable plan that translates the goals of the business plan into an appropriate facility response. To assist the Contractor in developing the FMP, the client portfolio is attached in Addendum to Technical Specifications for Building Facility Management & Maintenance: Lampiran 7A1 – Maklumat Asas Agensi Pelanggan.

- 6.6 The FMP shall include but not limited to planning, work/service methodology, manuals, resources, standard operating procedures (SOP), flow charts and responsibility matrix to produce the best result for each service. It should not be a lengthy document with wordy explanation but rather a precise document clarifying how to execute the service.
- 6.7 The Contractor shall manage and administer the services in accordance with the FMP that have been accepted by the FSO.
- 6.8 No review of, comments upon, acceptance of, or notice in respect of, any FMP or any other act or omission by the FSO about any management plan shall lessen or otherwise affect:
- a) the Contractor's liabilities or responsibilities under the Contract or otherwise according to law; or
 - b) the FSO's rights against the Contractor, whether under this Contract or otherwise according to law.
- 6.9 The Contractor shall submit the FMP within fourteen (14) days from the Commencement date. The Contractor shall submit improvised revision of FMP throughout the contract period as and when required by the FSO.
- 6.10 The FMP that shall be provided, but not limited to the following:
- a) FMM Work Program
 - b) Monthly Payment Plan
 - c) Human Resource Management Plan
 - d) Quality Management Plan
 - e) Materials, Spares, Tools & Equipment Management Plan
 - f) Transition Management Plan

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
		Section Revision: 2
SECTION B:	FACILITY MANAGEMENT PLAN (FMP)	Page 19 of 85

- g) Customer Care Management Plan
- h) Management Information System (MIS) Plan
- i) Utilities Management Plan
- j) Facility Risk Management Plan
- k) Facility Health, Safety and Environmental Management (HSE) Plan
- l) Incident Response and Disaster Recovery Management (IRDRM) Plan
- m) Energy Management Plan
- n) BIM Execution Plan Asset and Facility Management (BEPAFM)

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
SECTION C:	CONTRACTOR SITE ORGANISATION	Section Revision: 2
		Page 20 of 85

SECTION C: CONTRACTOR SITE ORGANISATION

7.0 FMM WORK PROGRAM

- 7.1 The FMM Work Program should cover all aspects of:
- Facility Engineering Operation & Maintenance Services and
 - Facility Custodial Services.
- 7.2 The Contractor shall provide information in the FMM Work Program such as (but not limited to) flow charts, graphical representation of predicted tasks, milestones, dependencies, resource requirements, responsibility matrix, task duration and deadlines for the FMM Work Program
- 7.3 The FMM Work Program shall be detailed enough to show each activities and services breakdown structure task to be performed, the position / designation of the person responsible for completing the task, the start and end date of each task, and the expected duration of the task.
- 7.4 The purpose of the FMM Work Program is to provide a useful 'road map' that can be used by the Contractor to:
- Plan, monitor, and control the timely execution of the activities and services, from the Commencement Date until the end of this contract;
 - Track and monitor the progress of the activities and services;
 - Manage resources more smoothly;
 - Identify and monitor dependencies and constraints between services to avert preventable delays; and
 - Communicate more frequently and effectively with FSO and user/customer.
- 7.5 The Contractor shall submit a quarterly performance analysis report with regards to each Performance Indicator (PI).

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
		Section Revision: 2
SECTION C:	CONTRACTOR SITE ORGANISATION	Page 21 of 85

7.6 The Contractor shall submit the following reports at the frequency stated but not limited to Table 2 below:

NO.	NAME OF REPORT	ITEM	FREQUENCY	REMARKS
1	Management Organisation Structure	9.1	Once Only	Update when required
2	Facility Familiarisation Verification (FFV) report	13.1.5, 13.3.3	Once only	During Transition Period
3	Facility Condition Assessment (FCA) report	13.2.4	Once only	4 Months before contract End or as and when required
4	Energy Audits report	20.6, 20.8	Once only	3 Month prior to Commencement Date
5	Defect rectification work progress report	13.1.6, 13.2.5	Weekly	During Transition Period/Transition Coordination Group meeting
6	Monthly Performance Review Report		Monthly	
	6.1 Statutory Compliance report	5.8		
	6.2 Monthly Monitoring report	14.1.9, 22.1.6		
	6.3 Inspection, repair and replacement record report	23.1.2		
7	Technical Advice and Expert Proposal Report	26.1	Monthly or as when required	
8	Customer Satisfaction Survey Report	14.1.17	Monthly	
9	Performance analysis	7.5	Quarterly	With regards to each Performance Indicator

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
		Section Revision: 2
SECTION C:	CONTRACTOR SITE ORGANISATION	Page 22 of 85

10	Facility Risk Management Report	17.6	Quarterly	Update Risk Register
11	Health Safety & Environment (HSE) Audit Report	18.11c	Yearly	
12	Spare Parts audits report	11.4	Yearly	
13	Structure Integrity report	23.6.7	Once Only	
14	Incident Report	19.8	When required	
15	BIM data and CDE Contents	22.2.7	Every 6 months	

Table 2: List of reports

7.7 If the Contractor failed to submit any report on time (that had been agreed by the FSO), the Government shall have the right to deduct from any monies due or becoming due to the Contractor under the Contract or from the Performance Bond or as a debt due from the Contractor.

8.0 MONTHLY PAYMENT PLAN (MPP)

8.1 The Contractor shall provide a Monthly Payment Plan (MPP) of the services to enable effective financial projection for the whole contract period.

8.2 The Contractor shall develop and implement MPP that includes the projection of cash flow for the whole contract period. The MPP shall consist of the estimated cost for each month's activities and services in accordance with this Contract.

8.3 The Contractor shall provide the FSO with budget forecasts as accurate as possible and in a timely manner.

8.4 The Contractor shall use the MPP as a reference and guidance to the actual monthly payment.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
		Section Revision: 2
SECTION C:	CONTRACTOR SITE ORGANISATION	Page 23 of 85

- 8.5 The Contractor shall provide the FSO and her nominated agent access to any of the Contractor's financial records and reports that relate to the services at any time required by the FSO.
- 8.6 The submission of the monthly payment shall be based on the executed FMM activities and services of the Evaluation month.
- 8.7 Regardless of the acceptance of the FSO on the MPP, the actual monthly payment shall be based on actual implementation of FMM activities and services done as per assessment by FSO.
- 8.8 All documentation for the purpose of monthly payment shall be submitted seven (7) days after cut-off date.

9.0 HUMAN RESOURCE REQUIREMENT

9.1 Staffing / Staff Requirement

- a) The Contractor shall develop and implement a Human Resource Management Plan that includes, but is not limited to:
 - i. a detailed description of the Contractor's strategic recruitment policies;
 - ii. details of the Contractor's recruitment process for all types of staff including recruited foreign workers, expatriate and local staff;
 - iii. details of the pre-requisite experience and qualifications required for each staff member for each role;
 - iv. details of the processes that shall be employed by the Contractor to ensure that all staff are appropriately experienced and qualified to undertake their specific duties; and
 - v. details of skills improvement or on-job-training list as a part of the enhancement in skills and competencies for the staff.
- b) The Contractor shall take appropriate measures to ensure that key management personnel nominated in its accepted management plans hold the nominated positions for the whole duration of the contract, unless otherwise instructed by the FSO.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
		Section Revision: 2
SECTION C:	CONTRACTOR SITE ORGANISATION	Page 24 of 85

- c) The Contractor shall provide adequate numbers of staff to undertake the services from the Commencement Date until the date of contract end.
- d) The Contractor shall take appropriate measures to ensure that all staffs are suitably qualified, have a positive attitude, are highly motivated, competent, skilled and experienced relevant in the industry to undertake their respective duties.
- e) The Contractor shall only allow staff to perform a duty if they have successfully completed the relevant training for that asset.
- f) At all times Contractor's staff shall wear appropriate and suitable uniforms (provided by the Contractor) that have been accepted by the FSO.
- g) The Contractor shall adequately provide appropriate tools and equipment for all staff to successfully perform their duties.
- h) The Contractor shall conduct measures to ensure that the staffs are courteous, helpful and professional when dealing with user/customer and the FSO's personnel at all times.
- i) The Contractor shall ensure that all staff are competent, knowledgeable and well trained to deliver all the FMM Services.
- j) The Contractor shall develop and implement a Human Resource Management Organisation chart, subject to FSO's acceptance, that includes organisation structure, job description, responsibilities matrix, manpower deployment plan and lines of communication. The Contractor shall manage and update Human Resource Management Plan when necessary.
- k) Human Resource Management Organisation chart is suggested to have three (3) tier structure for effective and efficient operation namely:
 - i. Management – FM, Engineers
 - ii. Supervisory – Officers, Supervisors, Chargeman
 - iii. Operational – Clerks, Technician, Operatives, Sub-contractors

9.2 Competency of Operation and Maintenance Staff

- a) The Contractor shall ensure that the employed staffs have knowledge and competency needed to operate and maintain the facility.
- b) The Contractor shall regularly demonstrate continuous development program that are essential for the staff to improve their skills and

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	SECTION C: CONTRACTOR SITE ORGANISATION	Section Revision: 2 Page 25 of 85

competency in delivering the service such as conducting on-job-training or other skills enhancement training program.

c) The Contractor is required to employ adequate number of the following categories for the FMM Services as stated in the Contract but not limited to the following: -

- i. **Facility Manager** – as defined and certified by Governing Bodies and accepted by FSO (based on contract requirements)
- ii. **Facility Executive** – as defined and certified by Governing Bodies and accepted by FSO (based on contract requirements)
- iii. **Facility Supervisor** – as defined and certified by Governing Bodies and accepted by FSO (based on contract requirements)
- iv. **Competent Electrical Engineer** - as defined and certified by Suruhanjaya Tenaga (visiting only - based on contract requirements)
- v. **Mechanical Engineer** - Degree in Mechanical Engineering and registered under Board of Engineer Malaysia (BEM) accepted by FSO (based on contract requirements)
- vi. **Electrical Engineer** - Degree in Electrical Engineering and registered under Board of Engineer Malaysia (BEM) accepted by FSO (based on contract requirements)
- vii. **Civil Engineer** - Degree in Civil Engineering and registered under Board of Engineer Malaysia (BEM) accepted by FSO (based on contract requirements)
- viii. **Facility Management Inspector** – as defined and certified by Governing Bodies and accepted by FSO (based on contract requirements)
- ix. **Competent Electrical Chargeman** - as defined and certified by Suruhanjaya Tenaga with at least 3 years of experience (based on contract requirements)
- x. **Competent Electrical Wireman** - as defined and certified by Suruhanjaya Tenaga with at least 3 years of experience (based on contract requirements)

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	SECTION C: CONTRACTOR SITE ORGANISATION	Section Revision: 2 Page 26 of 85

- xi. **Competent Boilerman** - as defined and certified by Jabatan Keselamatan dan Kesihatan Pekerjaan (JKKP) (based on contract requirements)
- xii. **Mechanical Supervisor** – Degree / Diploma in Mechanical Engineering and preferably certified by Governing Bodies and accepted by FSO and with at least 3 years of experience in maintenance of mechanical services.
- xiii. **Electrical Supervisor** – Degree / Diploma in Electrical Engineering or certified Chargeman and preferably certified by Governing Bodies and accepted by FSO and with at least 3 years of experience in maintenance of electrical services.
- xiv. **Civil Supervisor** - Degree/Diploma in Civil Engineering and preferably certified by Governing Bodies and accepted by FSO and with at least 3 years of experience in maintenance of civil services.
- xv. **Technicians Mechanical/Electrical/Civil** - with at least 3 years of experience for certificate holder from polytechnic/ SPVM.
- xvi. **General Worker/Handyman/Plumber** – with at least 3 years of relevant experience in Building Operation & Maintenance Services.
- xvii. **Quality Officer** as defined and certified by Authorised Bodies and accepted by FSO with 3 years of experience in Quality Management. (based on contract requirements)
- xviii. **Site Safety Supervisor (SSS)** as defined and certified by *Jabatan* Keselamatan dan Kesihatan Pekerjaan (JKKP) and accepted by FSO (based on contract requirements)
- xix. **Safety & Health Officer (SHO)** - as defined and certified by Jabatan Keselamatan dan Kesihatan Pekerjaan (JKKP) and accepted by FSO (based on contract requirements)
- xx. **Competent Scaffolder** - as defined and certified by Jabatan Keselamatan dan Kesihatan Pekerjaan (JKKP) and accepted by FSO (based on contract requirements)
- xxi. **Energy Officer** - as defined and certified by Governing Bodies approved by Government/Suruhanjaya Tenaga with minimum 3 years of experience in Energy Management. (based on contract requirements)

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
		Section Revision: 2
SECTION C:	CONTRACTOR SITE ORGANISATION	Page 27 of 85

- xxii. **Energy Manager** - as defined and certified by Governing Bodies approved by Government/Suruhanjaya Tenaga with minimum 3 years of experience in Energy Management. (based on contract requirements)
- xxiii. **Radiation Protection Officer (RPO)** - as defined and certified by Lembaga Perlesenan Tenaga Atom (AELB) (visiting only - based on contract requirements)
- xxiv. **Information and Communication Technology Personnel** – Degree/Diploma/Certificate in Information and Communication Technology Engineering or Science (based on contract requirements)
- xxv. **Site Administrative Personnel** - Degree/Diploma/Certificate in Business Administration / Finance / Account / ICT / Human Resource / Management or equivalent (based on contract requirements)
- xxvi. **BIM Modeller** – Diploma in a field related to Building Information Modelling (BIM) / Malaysian Skills Certificate (Sijil Kemahiran Malaysia, SKM) Level 4 in BIM / any related Diploma with certified Sijil Kecekapan Kemahiran Personel, SKKP in BIM by CIDB or equivalent certification (based on contract requirements)

9.3 Working Hours

- a) The Contractor shall submit the proposed working hours of the staff for all services to the FSO for approval. The Contractor shall submit revised working hours of the staff whenever required as per Lampiran 7A1 - Maklumat Asas Agensi Pelanggan.
- b) If the building complex is equipped with Building Control System (BCS), the BCS Room shall be manned 24 hours and shall not be left unattended at any time. BCS technician shall not be involved in corrective or preventive maintenance work outside the room.
- c) The Contractor shall be required to have a minimum competent number of staff which will be determined by the FSO based on the site condition on weekend and Public Holiday.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
		Section Revision: 2
SECTION C:	CONTRACTOR SITE ORGANISATION	Page 28 of 85

- d) At all other time (outside the working hours above), the Contractor's staff shall be on call and shall respond immediately when required.
- e) The Contractor's staff shall also be available for standby duties during functions and special occasions if there are request from Client.

9.4 Safety, Health and Environment (SHE) of the Staff

- a) The Contractor shall be required to provide a complete First Aid Kit as stated under Factory and Machinery (Safety, Health and Welfare) Regulation 1970 which shall be kept and properly maintained in the Contractor's site office. The kit shall be in the charge of either the Contractor's site representative or some other responsible person who shall be on the Site during all working hours to ensure that the first aid facility are available without delay at all times when work is in progress. At least one (1) designated person of the Contractor's staff shall be trained in first aid duties.
- b) The Contractor shall refrain from dumping and/or depositing any form of materials that can collect water which afford breeding places for mosquitoes, rodents, insects and vermin of any kind. All excavation and any portion of the site where water stagnates or accumulates shall be kept dry by pumping, bailing or other operations. The Contractor shall pay all charges as may be required by the Ministry of Health (MOH) and/or Local Authority and employ whatever destructive measures as are necessary.
- c) All works shall be carried out without unreasonable noise level and dust emission. The Contractor shall take measures to ensure that all equipment and machinery are in proper working condition to minimize the amount of noise and dust generated. The FSO may require the Contractor to submit a proposal on how to reduce excessive noise and dust.
- d) The Contractor is prohibited from discharging oil and grease to any water course. Storage tanks for oil and grease shall be placed on concrete base with up stand edges to contain any spillage. Any spilled oil and grease shall be promptly removed by the Contractor. The Contractor shall collect and store used oil, grease and other scheduled wastes and dispose these according to methods accepted by Department of Environment (DOE) and related Acts.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
		Section Revision: 2
SECTION C:	CONTRACTOR SITE ORGANISATION	Page 29 of 85

- e) All safety measures shall be carried out in accordance with Occupational Safety and Health Act (OSHA) 1994 and relevant local by-laws. The Contractor shall be held solely responsible for all accidents arising from any negligence in this respect.

10.0 QUALITY MANAGEMENT SYSTEM

The Contractor shall ensure that its management and operational processes and systems are reviewed in a coordinated and structured manner that allows any errors or non-conformances to be identified and appropriate corrective action plans implemented as soon as the non--conformance is detected.

10.1 Quality Management System

- a) The Contractor shall adopt a Quality Management System that includes non-conformance trigger points based on the operational, contractual and financial matters pertaining to the Contract.
- b) The Contractor shall ensure the accepted Quality Management Plan to be implemented and maintained to the latest version of MS ISO 9000, shall be submitted within fourteen (14) days from the Commencement Date and shall be maintained throughout the contract period.
- c) All quality system records and all records relating to the quality of the services shall be freely accessible to the FSO.
- d) The Contractor shall provide all necessary and appropriate information, facilities and cooperation during any inspection, investigation and auditing conducted by the Government or any officer or person authorized in writing by the Government at any time during the Contract Period.
- e) The Contractor shall develop and provide the Quality Management Plan that includes, but is not limited to:
 - i. define the responsibility and authority and reporting function of personnel primarily responsible for upholding the quality provisions of the contract;
 - ii. identify how independent inspection, witnessing and monitoring and reporting shall be carried out;
 - iii. identify the qualifications, experience and required competencies of personnel who shall undertake the duties required.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
		Section Revision: 2
SECTION C:	CONTRACTOR SITE ORGANISATION	Page 30 of 85

- iv. describe how the outcome of user/customer satisfaction shall be measured and initiate trigger points for improvement processes to be initiated;
 - v. describe how variation of service shall be measured and how large variations shall initiate improvement processes in the Contractor's management processes;
 - vi. address the safety in every aspect of the services;
 - vii. prepare the Standard Operating Procedures (SOP) documents for all processes in FMM Services.
- f) The Contractor shall comply with Sistem Pengurusan Bersepadu (SPB) JKR as specified by FSO.

11.0 MATERIALS, SPARES, TOOLS & EQUIPMENT REQUIREMENT

- 11.1 The Contractor shall strategize and adopt the best practices for the materials, spares, tools & equipment management of the building asset/equipment to ensure that there shall be minimum waiting time for operation and maintenance works.
- 11.2 The Contractor shall ensure a computerised database of all materials, spares, tools & equipment to be included as a module in Management Information System (MIS). The Contractor shall manage and maintain the database throughout the contract period.
- 11.3 The Contractor shall assess and identify the 'fast moving' and critical item to ensure that there is always sufficiently available stock of spare parts at all times during the operation period. For the spare parts that do not need to be in stock at the Site, the Contractor shall elaborate the process of getting the parts in the Material & Spares Management Plan to minimize downtime.
- 11.4 The Contractor shall undertake annual audits of the above materials, spares, tools & equipment or as and when directed by the FSO.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
SECTION C:	CONTRACTOR SITE ORGANISATION	Section Revision: 2
		Page 31 of 85

11.5 FSO has the right to audit materials, spares, tools & equipment inventory at any time without prior notice to the Contractor.

12.0 CONTRACTOR SITE OFFICE, INSPECTION TOOLS, EQUIPMENT & FACILITIES

12.1 The Contractor shall provide and maintain a site office and other facilities as below within or adjacent to the Site:

- a) Administration office
- b) Help desk office
- c) Meeting/discussion room
- d) Technical library
- e) Store room

12.2 The Contractor shall provide all necessary furniture, office tools and other facilities for the staff to function effectively.

12.3 The Contractor shall provide and maintain unit of computer workstation dedicated for Building Information Modelling (BIM) works. The workstation shall comply with the latest hardware specifications recommended for BIM applications and be equipped with approved BIM software as endorsed by FSO. The Contractor shall ensure that all BIM-related software licences are valid and updated throughout the contract period.

12.4 The Contractor shall provide all the required tools, equipment, and facilities for the use of the F.S.O and his staff throughout the contract period for monitoring, inspection, investigation, and auditing of FMM Services. The Contractor shall ensure that all necessary tools and equipment are in good condition, well maintained, calibrated, and available for use whenever required.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
SECTION D:	FACILITY MANAGEMENT SERVICES	Section Revision: 2
		Page 32 of 85

SECTION D: FACILITY MANAGEMENT SERVICES

13.0 TRANSITION MANAGEMENT

The Contractor shall develop and submit the Transition Management Plan for FSO acceptance as below:

- i. Transition-In Management Plan
- ii. Transition-Out Management Plan

13.1 Transition-In Management Plan

- 13.1.1 Transition-In period means the maximum period of three (3) months from Commencement Date to date of operation of FMM Services
- 13.1.2 The FSO shall convene a Transition Coordination Group that includes members from the FSO's office, the Contractor, successor Contractor and client.
- 13.1.3 The FSO shall initiate and facilitate monthly or as and when required meetings of the Transition Coordination Group.
- 13.1.4 The successor Contractor shall manage the transition activities in a coordinated and structured manner.
- 13.1.5 The successor Contractor shall carry out the Facility Familiarization Verification (FFV) including verification of Facility Condition Assessment (FCA) and submit the report to the FSO.
- 13.1.6 The successor Contractor shall verify the rectification work done by the Contractor or relevant parties and report to the FSO on weekly basis.
- 13.1.7 The successor Contractor shall identify new defects which not includes in FCA and immediately notify the FSO.
- 13.1.8 The successor Contractor shall provide appropriate personnel to attend and witness testing and verification and all handover activities to be undertaken by the appropriate parties including all trainings and activities required by FSO.
- 13.1.9 The successor Contractor shall alert and notify if there are any FMM activities that was not properly implemented accordingly to the Contractor's scope of work to the FSO.
- 13.1.10 The successor Contractor shall submit proposed list of key management personnel and site transition team to be accepted by FSO.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
SECTION D:	FACILITY MANAGEMENT SERVICES	Section Revision: 2
		Page 33 of 85

13.1.11 The successor Contractor shall take over the services relating to any commissioned utility or services at the request of the FSO.

13.1.12 The Transition-In Management Plan shall include but is not limited to:

- a) Organisational charts including key management personnel and site transition team;
- b) Schedules of activities during transition management;
- c) Communication chart, roles and responsibilities;
- d) Standard operating procedure and method of statement to manage the services to ensure the requirements are met and details of how the successor Contractor shall manage or assist any other relevant services such as:
 - i. on-going construction works
 - ii. any remaining Defect Liability item from construction

13.2 Transition-Out Management Plan

13.2.1 Transition-Out means the period of three (3) months before the expiry of the FMM Contract Period.

13.2.2 The Contractor shall revise Transition-Out Management Plan and resubmit no later than one (1) month before the start date of Transition-Out.

13.2.3 The Contractor shall facilitate and co-operate with the successor Contractor for the Facility Familiarisation and Verification (FFV) activities.

13.2.4 The Contractor shall undertake the final FCA and submit the report to the FSO 4 months before contract end.

13.2.5 The Contractor shall ensure all defects identified in the final FCA report are rectified accordingly and report to the FSO on monthly basis.

13.2.6 If the rectification work of defect or any outstanding corrective work done by the Contractor is unaccomplished until the end of contract, the FSO have the right to deduct any monies due or becoming due to the Contractor under the FMCM or from the Performance Bond or as a debt due from the Contractor.

13.2.7 The Contractor shall carry out testing and verification and all handover activities.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	SECTION D: FACILITY MANAGEMENT SERVICES	Section Revision: 2 Page 34 of 85

13.2.8 The Contractor shall provide sufficiently appropriate personnel to attend to any activities required by the FSO.

13.2.9 The Contractor shall ensure that all scope of FMM Services are implemented and the appropriate staffs are deployed during transition period.

13.3 Facility Familiarization and Verification (FFV)

13.3.1 The Contractor shall provide training of trainers for successor Contractor and FSO to familiarize on any technology application applied in the site during the transition period.

13.3.2 The Contractor shall contribute and participate with the FSO induction program or training for FSO, client and user/customers.

13.3.3 The training for FFV shall contain the following not limited to:

- a) Tasks and responsibilities for their job role;
- b) Building and operation familiarization for all FMM scope;
- c) Technologies or systems relevant to the provision of the services;
- d) Safety and health requirements
- e) Site access and security;

13.3.4 The Contractor shall allow the successor Contractor to verify all the site information and asset inventories during the transition period.

13.4 Facility Condition Assessment (FCA)

13.4.1 The Contractor shall carry out the FCA to evaluate the physical state/condition of building elements and services and to assess the maintenance needs of the facility. The Contractor shall carry out the FCA as per item 13.2.4 or as and when instructed by FSO. The FCA generally comprises:

- a) physical inspection of a building to assess the actual condition of the site/building and its individual elements and services (e.g. air conditioning, fire protection), in comparison to the asset owner's specified condition standard or Government requirement;
- b) identification of maintenance works required to bring the condition of the building and its services up to, or maintain it at the specified condition standard or Government requirement;

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	SECTION D: FACILITY MANAGEMENT SERVICES	Section Revision: 2 Page 35 of 85

- c) ranking of maintenance works in order of priority; and
- d) determination by the assessor of actions to mitigate any immediate risk until remedial works (or other actions) can be taken to address problems.

13.4.2 The Contractor's FCA shall conform to the latest Government's Circular, Standard, Regulation, references and refer to the latest edition of Garis Panduan Pemeriksaan Keadaan Fasilitas to carry out the Facility Condition Assessment.

13.4.3 During the Transition-In period, the successor Contractor shall carry out a joint inspection and condition assessment/survey of the asset and facilities based on the latest FCA report with the FSO, user/customers' representative and Construction Contractor / FMM Contractor to determine the condition and integrity of the asset and facilities before taking-over. The methodology used for the joint inspection and condition assessment must be agreed upon by the FSO.

13.4.4 The Construction Contractor / FMM Contractor shall submit the above assessment report to FSO and sign off between all parties involved with the acceptance of all assets and facilities at the end of Transition-In period.

13.4.5 The Contractor shall submit the FCA report to FSO within the specific time as required by FSO.

13.5 Handover of the Asset

13.5.1 New FMM Contract

The Contractor shall organise joint inspection / auditing with the asset owner and FSO to determine the availability / functionality of all the asset services. An agreed inventory and defect list are to be produced and submitted to all parties involved (asset owner, FSO & Construction Contractor).

13.5.2 Existing FMM Contract:

A final joint inspection shall be carried out four (4) weeks prior to completion of this contract to verify the defects rectification works which specified in the final FCA report. Any outstanding works shall be executed by a third party and payment may be deducted from any

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	SECTION D: FACILITY MANAGEMENT SERVICES	Section Revision: 2 Page 36 of 85

monies due to the Contractor under this contract subject to FSO jurisdiction.

- 13.5.3 It is compulsory for the Contractor to handover all the systems in good working order to the successor Contractor and shall be properly documented.
- 13.5.4 All the MIS item related with Management, Operation and Maintenance Systems / Data / Records / Manuals / As-built Drawings / Asset Register / Inventory shall be updated to the CMMS.
- 13.5.5 The Contractor shall hand over any Government owned spare parts left to the successor Contractor.

14.0 CUSTOMER CARE MANAGEMENT

- a) The Contractor shall be committed to meet the needs and expectation of the user/customer and shall be responsible for overall management of Customer Care Service.
 - b) The Contractor shall develop and provide a Customer Care Management Plan that includes, but is not limited to:
 - i. organisational charts and communication plan complete with name, photo, designation etc.;
 - ii. escalation process plan (internal and external);
 - iii. planned resourcing by location considering the phasing and anticipated demographics;
 - iv. details of how the Contractor shall manage the service; and
 - v. Customer Satisfaction Survey (CSS) implementation strategy
 - c) The Contractor shall ensure that all staff at all levels interact with user/customer professionally and act courteously at all times when carrying out their respective duties.
- 14.1 Customer Care Service
- 14.1.1 The Contractor shall provide Customer Care Service 24 hours per day 365 days per year basis to address emergency, urgent and/or routine requests. The Contractor shall respond to such service requests within the service Response Time set out in the Performance Requirements

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	SECTION D: FACILITY MANAGEMENT SERVICES	Section Revision: 2 Page 37 of 85

and shall return the affected element(s) to the required quality standard within the allotted execution time. The Customer Care Services channel shall include but not limited to:

- a) Help Desk services
- b) 24 hours on-call services
- c) Any messaging and/or application system available

14.1.2 The Contractor shall ensure that:

- a) clear instructions for the user/customer on the agreed channel of the Customer Care Service shall be provided to the FSO at least one (1) month before the Customer Care Service commences operations (Help desk service to assume operation);
- b) updates to the instructions shall be provided to the FSO from time to time, as necessary;
- c) all user/customer, service providers, staff and public are familiar with these help desk service instructions.

14.1.3 The Contractor shall log all service requests made and events reported. The Contractor shall record all relevant details, including but not limited to the following information:

- a) requester's / reporter's name;
- b) date and time;
- c) location (functional area and unit);
- d) nature of the service request or event;
- e) service required;
- f) categorization (priority);
- g) expected response and execution time;
- h) unique request reference;
- i) staff name/contact to which the work request/order was passed;
- j) date and time the work request/order passed to;
- k) action taken; and
- l) actual response time and execution time achieved.

14.1.4 The Contractor shall generate an activity request report (Work Request) for each complaint or event reported. The activity request report shall be communicated to the relevant staff and/or service provider. Such communication shall be properly documented.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	SECTION D: FACILITY MANAGEMENT SERVICES	Section Revision: 2 Page 38 of 85

14.1.5 The Contractor shall provide standard signage 'Talian Bantuan/Aduan Kerosakan' consists of Government Logo together with the person in charge's name and contact number to be located at strategic location.

14.1.6 The Contractor shall inform the user/customer as to the proposed course of action and response time and execution time allocated. The help desk service shall coordinate the appropriate response to all complaints.

14.1.7 The Contractor shall respond to the complaints or event reported and on completion of the remedial activity shall inform the help desk service together with the achieved response, execution time and the action undertaken. This information has to be logged onto the computerised system.

14.1.8 The Contractor shall provide sufficient moveable signage and notify user/customer for any corrective or preventive maintenance works that directly affect user/customer using standard signage consists of (but not limited to) 'Kerja-kerja Pembaikan / Penyenggaraan/ Pembersihan sedang dijalankan, maaf di atas kesulitan yang berlaku', Work Order number, estimated date or time to finish and name and contact number of person in charge.

14.1.9 The Contractor shall provide all information and/or data as required to complete the Monthly Monitoring report. This report shall include but not be limited to:

- a) Help Desk Service status report as in item 22.1.6
- b) a summary of all complaints and events reported to the help desk during the contract month including allotted target response / execution times where applicable and those achieved;
- c) a summary of all failure events and quality failures;
- d) the functional part(s) / unit(s) affected; and
- e) the duration (in hour), of any failure event not rectified on time.

14.1.10 Information logged with the help desk shall not be amended unless there is a system in place to record:

- a) the exact nature and impact of the amendment;
- b) the reason for the amendment;
- c) by whom the amendment was authorized.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
		Section Revision: 2
SECTION D:	FACILITY MANAGEMENT SERVICES	Page 39 of 85

14.1.11 The FSO shall be given full access to all help desk records at any time.

14.1.12 In the event of an emergency, at whatever time, the help desk shall assist in escalation process by reporting the incident to internal and external authorities, coordinating the response, logging the details and raising the alarm (if any).

14.1.13 The Contractor shall keep the user/customer informed should problems occur in executing the service response.

14.1.14 The Contractor shall ensure confidentiality is maintained.

14.1.15 The Contractor shall carry out Customer Satisfaction Survey monthly to gauge the effectiveness and performance of its FMM Service delivery.

14.1.16 The Contractor shall prepare Customer Satisfaction Survey Reports that includes, but not limited to the following:

- a) demographics information;
- b) statistical information;
- c) individual subject analysis;
- d) analysis by department;
- e) analysis on services and systems provided;
- f) summary of analysis;
- g) analysis of each respondent;
- h) impact study; and
- i) Customer Satisfaction improvement action plan.

14.1.17 Where CSS is found to be below KPI target, the Contractor shall carry out a review and produce an action plan to address the problem

14.2 Help Desk Personnel

14.2.1 The Contractor shall provide adequately trained and qualified staffs to handle the Help Desk service.

14.2.2 The Contractor shall ensure that all Help Desk personnel are courteous, helpful and professional in their manner when interacting with user/customer/FSO's personnel at all times.

14.2.3 The Contractor shall ensure that all Help Desk personnel speak a good level of Bahasa Malaysia and a fair / reasonable command of English.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	SECTION D: FACILITY MANAGEMENT SERVICES	Section Revision: 2 Page 40 of 85

14.3 Liaison / Communication

- 14.3.1 The Contractor shall put in place appropriate policies, procedures and processes to ensure the implementation of effective liaison / lines of communication with the Client, Users/customers, FSO and other relevant external parties.
- 14.3.2 Client and user/customer's representatives shall be the line of communication for the Client and user/customer with the Contractor and FSO's representatives. The Contractor must ensure that the Client and user/customer's requirements, complaints and proposals are taken care of professionally, effectively and with the approval of FSO.
- 14.3.3 The Contractor shall attend FMM Coordination Meeting chaired by FSO to discuss matters relation to the contract.
- 14.3.4 The Contractor shall ensure to equip all the staff (including sub-contractors and service providers) with relevant and sufficient soft skills when dealing with the user/customer while performing the service.

15.0 MANAGEMENT INFORMATION SYSTEM (MIS)

15.1 Management Information System (MIS) Plan.

The Contractor shall prepare MIS Plan that include, but is not limited to:

- 15.1.1 an organisational chart describing Contractor personnel responsible for managing each part of the information systems;
- 15.1.2 details of the processes the Contractor shall undertake to manage the requirements of all information systems;
- 15.1.3 details of any communications protocols to be followed in respect of availability and reliability of the systems;
- 15.1.4 details of how the Contractor shall ensure the requirements of the scope of work shall be met with respect to the information systems accepted by the FSO;
- 15.1.5 details of measures and procedures that shall be implemented in the event of a failure or loss of any part of any of the systems being operated and maintained by the Contractor.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	SECTION D: FACILITY MANAGEMENT SERVICES	Section Revision: 2 Page 41 of 85

15.2 Application Management

- 15.2.1 The Contractor shall develop, implement and utilize a Management Information System (MIS) subject to FSO's acceptance as required in PWD Facilities Management & Maintenance Form (2016) Clause 22.
- 15.2.2 If there is an Existing MIS, the Contractor shall manage, maintain, amend, upgrade and update the Existing MIS accordingly.
- 15.2.3 Computerised Maintenance Management System (CMMS): Refer to Lampiran 7A2 - JKR Computerised Maintenance Management System (CMMS) Specification for detail specification and requirements or latest version.
- 15.2.4 Common Data Environment (CDE): Where Building Information Modelling (BIM) is implemented, the Contractor shall establish and maintain a Common Data Environment (CDE) as part of the Management Information System (MIS). The CDE shall be the official digital platform for storing, managing and sharing BIM data, models and related asset information. The Contractor shall ensure that the management and exchange of such information between the BIM authoring tools and the CMMS are clearly addressed in the MIS Plan, either through manual or other approved methods. The Contractor shall also ensure that the FSO and the Client are granted appropriate access to the Contractor's CDE platform for review, monitoring, and coordination purposes.

15.3 Technical Library & Repository

- 15.3.1 The Contractor shall locate, identify, label and register all assets and inventories within three (3) months from the Commencement date.
- 15.3.2 The Contractor shall develop, procure, manage and maintain a structured administration system for management of all facilities and maintenance-related information under the contract, to ensure transparency of process and full audit trails. The technical library shall be available for scrutiny by the FSO. The core components of the technical library shall consist of the following:
 - a) the technical documents and records (as-built drawings, O&M manuals etc.); and

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	SECTION D: FACILITY MANAGEMENT SERVICES	Section Revision: 2 Page 42 of 85

b) the asset register / inventory.

15.3.3 The Contractor shall initiate this process in 15.3.2 by reviewing and commenting where necessary the Operations and Maintenance (O&M) Manual. All technical documents shall be updated when required and upload in CMMS accordingly.

15.3.4 The Contractor shall maintain a true and correct set of records and a complete up to date and orderly documentary record of all transactions.

15.3.5 All documents shall be kept in good order and shall be available at all reasonable times for inspection by the FSO, and the Contractor shall make available such information as may be reasonably required by the FSO.

15.4 Repository

15.4.1 In the event the Contractor installs any new equipment or replaces any equipment, the Contractor shall supply four (4) sets of operation and maintenance manuals, and spare parts books for all such equipment to the FSO, together with four (4) sets of detailed and itemized list of spare parts which the manufacturers of the supplied equipment consider as essential to be kept in ready stock for the purpose of service and maintenance.

15.4.2 All the management plan, operation and maintenance manual / CMMS / data / records / manuals / as-built drawings / asset register / inventory shall be up-to-date and remain as the property of the Government.

15.4.3 The item in 15.4.2 shall be officially handed over by The Contractor to the FSO and the successor Contractor at the end of the contract or when the contract is terminated to ensure continuity in the management, operation and maintenance of the assets.

15.5 Warranty Management

15.5.1 The Contractor shall be primarily responsible for managing any maintenance works under the Contract, which shall be carried out by specialist technical Contractors procured by the Contractor.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	SECTION D: FACILITY MANAGEMENT SERVICES	Section Revision: 2 Page 43 of 85

15.5.2 The Contractor shall, in preparing and implementing the Maintenance Plan of new asset, ensure that the original equipment manufacturers' warranty agreements are not infringed.

16.0 UTILITIES MANAGEMENT

- a) The Contractor shall develop and implement a Utilities Management Plan that includes, but is not limited to:
- i. an organisational chart describing Contractor personnel responsible for managing the services for each utility;
 - ii. details of the processes the Contractor shall undertake to manage the requirements of each utility agreement;
 - iii. details of any communications protocols to be followed for liaising with utility providers and user/customer in respect of supply and distribution or reticulation of each service;
 - iv. details of how the Contractor shall ensure the requirements of all utilities shall be met during the transition-in and transition-out period;
 - v. details of measures and procedures that shall be implemented in the event of a failure or loss of any part of any of the utilities or services provided; and
 - vi. the steps the Contractor shall take to implement the requirements of the Utilities Management Plan into its management processes.

16.1 Electricity Supply

16.1.1 The Contractor shall manage the operation and maintenance aspects of the electricity distribution network excluding all matters relating to pricing, invoicing, billing and payment collection.

16.1.2 The key outcomes of successful management of the operation and maintenance aspects of the electricity supply agreement shall be reliable and continuous electricity supply to all user/customer and the reliable metering of each user/customer' consumption.

16.1.3 The Contractor shall undertake all of the operation and maintenance of the electricity distribution network system after the meter.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	SECTION D: FACILITY MANAGEMENT SERVICES	Section Revision: 2 Page 44 of 85

16.1.4 The Contractor shall accurately record and submit all electricity meter readings to the FSO. The Contractor shall inform FSO/client and liaise with service provider if there are faulty/problems in metering.

16.1.5 The Contractor shall dispatch any user/customer enquiries relating to electricity supply costs, charges, invoices or payments directly to the FSO/service provider. The Contractor shall handle all other enquiries relating to the supply and distribution of electricity.

16.1.6 The Contractor shall provide technical advice to the FSO in respect of electricity supply and distribution (as & when necessary).

16.2 Cold Water Supply

16.2.1 The Contractor shall manage the operation and maintenance aspects of the cold-water distribution network system excluding all matters relating to pricing, invoicing, billing and payment collection.

16.2.2 The key outcomes of successful management of the operation and maintenance aspects of the cold-water supply agreement shall be a reliable quality water supply to all user/customer at all times and the reliable and accurate metering of each user/customer's consumption.

16.2.3 The Contractor shall accurately record and submit all cold-water meter readings to the FSO. The Contractor shall inform FSO/client and liaise with service provider if there are faulty/problems in metering.

16.2.4 The Contractor shall be responsible for ensuring that water quality testing is regularly undertaken on the cold-water supply network to monitor the water quality provided to user/customer against the most current revision of the National Water Quality Standards.

16.2.5 In the event the cold-water quality does not meet the most recent revision of the National Water Quality Standards at the time of sampling, the Contractor shall immediately notify the FSO and recommend appropriate action in accordance with its Incident Response and Disaster Recovery Management Plan.

16.2.6 The Contractor shall dispatch any user/customer enquiries relating to cold water supply costs, charges, invoices or payments directly to the

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	SECTION D: FACILITY MANAGEMENT SERVICES	Section Revision: 2 Page 45 of 85

FSO/service provider. The Contractor shall be responsible for all other enquiries relating to the supply and distribution of cold water.

16.2.7 The Contractor shall provide technical advice to the FSO in respect of cold water supply and distribution (as & when necessary).

16.3 Chilled Water Supply (if applicable)

16.3.1 The Contractor shall accurately provide all chilled water meter readings to the FSO where applicable. The Contractor shall inform FSO/client and liaise with service provider if there are faulty/problems in metering.

16.3.2 The Contractor shall help manage the chilled water on behalf of the FSO.

16.3.3 The Contractor shall undertake all the operation and maintenance of the chilled water supply system and distribution network system in the secondary chilled water loop (demand side). The Contractor shall proactively liaise with service provider if there is any problem with primary chilled water loop (supply side inclusive of heat exchangers).

16.3.4 The Contractor shall be responsible for ensuring that water quality and temperature testing is regularly undertaken on the chilled water supply network to monitor the water quality and temperature provided against the standard specified.

16.3.5 The Contractor shall dispatch any user/customer enquiries relating to chilled water supply costs, charges, invoices or payments directly to the FSO/service provider. The Contractor shall be responsible for all other enquiries relating to the chilled water supply.

16.4 Treated Sewerage Effluent (TSE)

16.4.1 The Contractor shall manage the operation and maintenance aspects of the (TSE) distribution network excluding all matters relating to pricing, invoicing, billing and payment collection.

16.4.2 The Contractor shall undertake all of the operation and maintenance of the TSE system.

16.4.3 The Contractor shall manage all aspects of its TSE management with service provider and the user/customer in respect of invoicing, billing and payment collection.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	SECTION D: FACILITY MANAGEMENT SERVICES	Section Revision: 2 Page 46 of 85

16.4.4 The Contractor shall dispatch any user/customer enquiries relating to TSE supply costs, charges, invoices or payments directly to the FSO/service provider. The Contractor shall be responsible for all other enquiries relating to the TSE system.

16.4.5 The Contractor shall provide technical advice to the FSO in respect of TSE system.

16.5 Sewerage Collection and Disposal Service

16.5.1 The Contractor shall manage the operation and maintenance aspects of the sewerage collection and disposal network.

16.5.2 The Contractor shall undertake all of the operation and maintenance of the sewerage system.

16.5.3 The Contractor shall manage all aspects of its management with the service provider and the user/customer in respect of invoicing, billing, payment collection and the master community service charge.

16.5.4 The Contractor shall dispatch any user/customer enquiries relating to sewerage costs, master community service charges, invoices or payments directly to the FSO/service provider. The Contractor shall be responsible for all other enquiries relating to the sewer system.

16.5.5 The Contractor shall provide technical advice to the FSO in respect of the sewerage network and sewerage disposal.

16.6 LPG / CNG Supply

16.6.1 The Contractor shall manage the operation and maintenance aspects of the LPG / CNG distribution network excluding all commercial matters including those relating to pricing, invoicing, billing and payment collection.

16.6.2 The Contractor shall undertake all of the operation and maintenance of the LPG / CNG supply and distribution network system after the bulk meter.

16.6.3 The Contractor shall accurately provide all LPG / CNG meter readings to the FSO. The Contractor shall inform FSO/client and liaise with service provider if there are faulty/problems in metering.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	SECTION D: FACILITY MANAGEMENT SERVICES	Section Revision: 2 Page 47 of 85

16.6.4 The Contractor shall manage all aspects its LPG / CNG supply management with the supplier and the user/customer in respect of invoicing, billing and payment collection.

16.6.5 The Contractor shall dispatch any user/customer enquiries relating to LPG / CNG supply costs, charges, invoices or payments directly to the FSO/service provider. The Contractor shall handle all other enquiries relating to the supply and distribution of LPG / CNG.

16.6.6 The Contractor shall provide technical advice to the FSO in respect of the LPG / CNG supply and distribution.

17.0 FACILITY RISK MANAGEMENT

17.1 The Contractor shall develop and implement a Facility Risk Management Plan that includes, but is not limited to:

- a) a Risk Register developed by the Contractor or the FSO;
- b) mitigation actions for each risk identified;
- c) responsible personnel for implementing and managing the mitigation measures for each risk;
- d) the process to implement the requirements of the Facility Risk Management Plan; and
- e) to conduct a discussion/ workshop among client/ stakeholder/ Contractor/ predecessor Contractor to finalise Facility Risk Management Plan and Risk Register.

17.2 The Contractor shall ensure that all risks are identified, effectively mitigated and managed through a measured process with involvement from various parties as accepted by FSO.

17.3 During the contract Period the Contractor shall review, identify, measure and register, in consultation with the FSO all risks that relate to the successful delivery of the services.

17.4 The Contractor shall document and implement effective measures to mitigate against each risk identified by the Contractor and the FSO.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
		Section Revision: 2
SECTION D:	FACILITY MANAGEMENT SERVICES	Page 48 of 85

17.5 The Contractor shall assign the responsibility to the appropriate personnel with the authority to manage the risk mitigation measures agreed and adopted by the Contractor.

17.6 The Contractor shall produce Facility Risk Management report, review and update its risk database at least every three (3) months with the FSO. The Contractor shall implement and assign any additional risks mitigation measures identified or improve any ineffective or inappropriate mitigation measures.

17.7 The Contractor's Facility Risk Management Plan shall conform to the latest Government's Circular, Standard, Regulation, references and refer to the latest edition of Garis Panduan Pengurusan Risiko Fasilitas and other risk related document to carry out the Risk Management.

18.0 FACILITIES HEALTH, SAFETY AND ENVIRONMENTAL (HSE) MANAGEMENT

18.1 The Contractor shall ensure and responsible that users/customers are always safe to use the asset and facilities in the Site.

18.2 The Contractor shall ensure that it develops, promotes and maintains a strong culture of health and safety in its workforce.

18.3 The Contractor shall incorporate health and safety in all aspects of the services including but not limited to the health and safety of all user/customer.

18.4 The Contractor shall manage the site / sites to minimise harm and prevent injuries or fatalities to user/customer.

18.5 HSE Management Plan shall also be developed and implemented as a component of the services during the transition period.

18.6 The Contractor shall incorporate Safety & Health Management System and Environmental Management System (EMS) aspect in HSE Management Plan.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
		Section Revision: 2
SECTION D:	FACILITY MANAGEMENT SERVICES	Page 49 of 85

18.7 The Contractor shall carry out all safety measures in accordance with Occupational Safety and Health Act (OSHA) 1994, Factories and Machineries Act 1967, Environmental Quality Act 1974, relevant regulations and local by-laws.

18.8 The Contractor shall be held solely responsible for all accidents arising from any negligence (includes negligence from the sub-contractors and service providers engaged in the premise) in performing FMM Services.

18.9 The Contractor shall establish Permit to Work (PTW) System for all the sub-contractors and service providers appointed internally and/or externally (by FSO/Clients) who will do work/ provide service within the Site. The Contractor shall issue a Stop Work Order or advice FSO/Client for the issuance of Stop Work Order should the appointed sub-contractors and service providers fail to establish / follow PTW process. PTW System shall be elaborated in details in HSE Plan.

18.10 The Contractor shall employ throughout the entire contract period a competent and qualified person as Safety and Health Officer (SHO) or Site Safety Supervisor (SSS) depending on the latest Safety & Health act, law and regulations. The SHO or SSS shall be responsible for liaising with the FSO and other engineers/officers in respect of all matters relating to safety and health.

18.11 The Contractor shall:

- a) comply with the accepted HSE Management Plan;
- b) permit representatives of the FSO to be present during such audits;
- c) deliver two (2) copies of yearly audit report to the FSO within ten (10) working days of its completion; and
- d) immediately implement corrective actions to close out all non-conformances contained in the audit report.

18.12 The Contractor shall develop and implement an HSE Management Plan that includes, but is not limited to the following:

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
		Section Revision: 2
SECTION D:	FACILITY MANAGEMENT SERVICES	Page 50 of 85

- a) details of how the Contractor shall manage the services to ensure the requirements of the HSE requirement are always met;
- b) details of how the Contractor shall ensure a seamless HSE interface with the current Contractor working concurrently during the Transition Period;
- c) details of how the Contractor shall address the following issues:
 - i. Hazard Identification, Risk Assessment and Risk Control (HIRARC);
 - ii. Induction/Safety Briefing/toolbox meeting to all workers (including sub-contractors/service providers before they commence on site;
 - iii. work processes for safe systems of work;
 - iv. Permit to Work (PTW) System;
 - v. accident and incident reporting including corrective action; and
 - vi. monthly team meetings to address health, safety and environmental issues.
- d) assessment of the risk, provide protection from and provide a remedy for any adverse environmental effect which may result from the performance of any component of the services;
- e) definition of the environmental responsibilities of the Contractor and those of each position within the Contractor's management team;
- f) inclusion of schedules of available resources, including personnel to deal with environmental incidents;
- g) the need for environmental safeguards and the adoption of environmentally sensitive work practices during any of the services including, but not limited to, procedures for:
 - i. Noise and air quality as per Indoor Environmental Quality (IEQ) Standard
 - ii. Water, vibration and groundwater monitoring and control;
 - iii. reducing noise levels;
 - iv. avoiding, reducing, reusing and recycling waste;
 - v. dust control including monitoring, mitigation and remedial actions;
 - vi. detection, treatment and disposal of contaminated materials and water;
 - vii. water quality control;
 - viii. erosion and sediment control;

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
		Section Revision: 2
SECTION D:	FACILITY MANAGEMENT SERVICES	Page 51 of 85

- ix. road traffic management regarding how it affects community safety and amenity; and
- x. collect and store used oil, grease and other scheduled wastes and dispose these according to methods accepted by DOE.
- h) the steps the Contractor shall take to implement the requirements of the management plan into its management processes.

18.13 The Contractor shall develop and operate an Environmental Management System (EMS). The EMS shall include appropriate monitoring systems to ensure compliance with sustainability requirements, Government environmental policy, its own environmental policy and its management system.

18.14 The Contractor shall ensure that:

- a) all relevant records are auditable by external consultants and available to support MS ISO 14001 and MS ISO 45001;
- b) an audit of refrigerants is maintained, with any losses reported;
- c) all fire extinguisher discharges and losses are recorded;
- d) any used fluorescent tubes and batteries are sent for safe disposal or recycling;
- e) there is no venting of recoverable ozone depleting substances from any equipment used by the Contractor in the provision of the services, and that these substances are recovered for recycling or destruction using appropriate technology;
- f) in addition, the Contractor shall, encourage the Payment of light, power and other resources and the reduction of pollution; and
- g) The Contractor shall update and maintain the EMS and building environmental policy when required to comply with changes in the client's environmental policy and legislation.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
		Section Revision: 2
SECTION D:	FACILITY MANAGEMENT SERVICES	Page 52 of 85

19.0 INCIDENT RESPONSE AND DISASTER RECOVERY MANAGEMENT (IRDRM)

- 19.1 The Contractor shall prepare, manage and respond to any incident on the asset. IRDRM procedures shall be accepted by FSO.
- 19.2 The Contractor shall respond to all incidents and ensure the scene of the incident is made safe, undertake preliminary restoration of the damage, notify the appropriate parties (including emergency services agencies) and assist the appropriate emergency services agencies' personnel.
- 19.3 The Contractor shall provide and use incident response personnel who are trained and skilled in providing 'first-on-scene' response services. The incident response personnel shall be responsible for:
- notifying emergency services agencies;
 - making the scene of an incident safe;
 - administering first aid if required;
 - undertaking preliminary rectification works;
 - providing traffic and crowd management around the incident site; and
 - liaising with and supporting any emergency services agencies when they arrive at the scene.
- 19.4 The Contractor's incident response services shall include undertaking minor repairs to damaged infrastructure until the appropriate maintenance or emergency services agencies arrive at the scene. The undertaking of any such service by the Contractor's personnel shall not be allowed to impede the primary role of maintaining safe conditions.
- 19.5 Any actions taken by the Contractor in response to an incident shall be in accordance with the accepted IRDRM Plan and may include notification to the FSO and other emergency services agencies and / or other authorities for action and/or dispatching a service crew.
- 19.6 The incident response personnel shall temporarily make safe or isolate an incident site properly.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
		Section Revision: 2
SECTION D:	FACILITY MANAGEMENT SERVICES	Page 53 of 85

19.7 The Contractor shall develop an IRDRM Plan in accordance with the contract which shall address:

- a) standard operating procedure to manage IRDRM;
- b) requirements of emergency and incident response agencies;
- c) processes for response planning and responding to incidents;
- d) systems and standards for all necessary actions likely to arise from incidents requiring rapid response;
- e) responsibilities for response measures within the Contractor's team;
- f) current details of all resources (personnel, plant and materials) available to the Contractor for the purpose of responding to incidents;
- g) communications systems;
- h) protocols regarding the occurrence of incidents;
- i) monitoring and recording, including response times;
- j) emergency agencies liaison procedures;
- k) training inclusive of regular drill;
- l) safety;
- m) facility risk management;
- n) record keeping;
- o) the requirements of the FSO and all relevant emergency services agencies; and
- p) management procedures that shall be implemented in the event of a disaster occurring including processes to be followed for recovery from disasters including temporary relocation of some operations facilities if required.

19.8 The Contractor shall prepare Incident Report upon any incident/ accident/disaster event that happens within the site boundary when required/instructed by FSO/Client.

19.9 The Contractor shall plan and coordinate an Incident Response drill exercise on an annual basis in coordination with Clients and emergency services agencies. The type of activities to be included in the drill shall be determined in consultation with Clients, emergency services agencies and the FSO.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
		Section Revision: 2
SECTION D:	FACILITY MANAGEMENT SERVICES	Page 54 of 85

19.10 The Contractor shall review and update the IRDRM Plan at least annually to ensure continued compliance with the contract.

19.11 The Contractor shall plan and coordinate half yearly simulation exercises to simulate the readiness of the Contractor to immediately respond, mitigate, put under control and rectify any emergency / disaster on failure / breakdown of critical services such as electricity supply, water supply, pipe bursts and fire inside the buildings. The implementation shall have approval from Clients and FSO.

20.0 ENERGY MANAGEMENT

20.1 The Contractor shall provide as part of the Client's Energy Management System (EnMS) a continuous support service for monitoring and managing energy usage and conservation. Steps taken and opportunities identified to reduce energy consumption shall be provided as part of the required management reporting.

20.2 The Contractor shall establish Baseline Energy Performance and propose yearly Energy Reduction Target.

20.3 The Contractor shall operate a continuous improvement process on energy usage and shall review annually the building performance against current best practice, and institute an energy conservation program for the following year.

20.4 The Contractor shall also demonstrate innovation in the efficient use of energy and materials to provide a sustainable environment, with management of the building demonstrating forward thinking on efficiency issues.

20.5 The contractor shall prepare a monthly energy usage report and greenhouse gas emission report analysis

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
		Section Revision: 2
SECTION D:	FACILITY MANAGEMENT SERVICES	Page 55 of 85

- 20.6 Energy Audit shall be undertaken in the first year of the new contract as per FSO instruction.
- 20.7 If there is an existing energy audit report, the Contractor shall refer to the output of the previous report (validity of the report is one (1) year).
- 20.8 The audit report shall include benchmarking of the building's energy consumption. The Contractor shall identify energy saving measures proposal in the audit report. The energy audit report shall be accordance to (but not limited) Garis Panduan Audit Tenaga JKR Bagi Bangunan Kerajaan 2014 (GPAT) or its latest version, MS 1525:2014 Energy Efficiency and Use of Renewable Energy for Non-residential Building - Code of Practice or its latest version.
- 20.9 Energy saving measures proposal in Energy Audit report on shall be submitted to the FSO who then may implement wholly or partially, or not at all, the proposals through the Contractor or a third party. Upon the completion of the implementation of the proposals, the Contractor shall manage, operate and maintain the installed systems/equipment.
- 20.10 The Contractor shall prepare Energy Audit Report that includes, but not limited to:
- Executive Summary (major finding etc.);
 - Introduction (background, disclaimer, data source etc.);
 - Description of facility or building audited – numbers of floors, GFA, air-con area, type of usage, occupancy, hours of operation, age of building, etc.
 - Information on tenants in facility or building audited;
 - Historical Energy Consumption (including Analysis of electricity bills at least twelve (12) months summation of usage);
 - Baseline energy consumption and the methodology used to establish it;
 - Details of measurement instrumentation used (parameters monitored and duration of monitoring for each parameter);
 - Energy Apportioning;

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
		Section Revision: 2
SECTION D:	FACILITY MANAGEMENT SERVICES	Page 56 of 85

- i) Load profiles analysis using power logger equipment at all Main Switch Board for each block to record energy usage for at least 1 week working days;
- j) Analysis of Building Energy Indices [Building Energy Intensity Index (BEII), Air-Conditioning Energy Intensity Index (ACEII), Lighting Energy Intensity Index (LEII)];
- k) Data plots of performance of systems or equipment;
- l) Energy Savings Measures proposal (with cost and no cost measures);
- m) Analysis of Chiller COP and efficiency (kW/ton) for at least 1 week working days;
- n) Yearly Energy Reduction Target;
- o) Summary and Implementation Strategy; and
- p) References & Appendices.

20.11 The Contractor shall ensure the Audit Report is certified by Registered Electrical Energy Manager (REEM).

20.12 The Contractor shall take initiatives to implement any no cost measures proposed in the Energy Audit Report. However, the Contractor shall coordinate with the clients/user/customers and FSO before any implementation of no cost measures.

20.13 All the relevant building energy parameters shall be monitor before and after the energy saving measures are implemented. Energy Baseline shall be established for energy saving comparison.

20.14 The Contractor shall take an initiative to provide enough awareness to the staffs and users/customers and shall integrate best energy management practices in the daily operation and servicing of the building system.

20.15 The Contractor shall proactively involve and liaise with the Client's Energy Management Committee (if any) to ensure the continuity and sustainability of the energy management program.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
SECTION D:	FACILITY MANAGEMENT SERVICES	Section Revision: 2
		Page 57 of 85

20.16 The contractor shall be equipped with energy measuring instruments for daily usage such as (but not limited to):

- a) Three phase power loggers for conducting basic energy usage measure/study,
- b) Sling psychrometer,
- c) Lux meter

20.17 The contractor shall ensure all energy measuring instruments are in good working condition and have a valid calibration if they are to produce reliable and accurate readings.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
SECTION E:	FACILITY ENGINEERING OPERATION AND MAINTENANCE SERVICES	Section Revision: 2
		Page 58 of 85

SECTION E: FACILITY ENGINEERING OPERATION AND MAINTENANCE SERVICES

21.0 GENERAL

21.1 The following section details out the minimum service requirements for which the successful Contractor must undertake to provide a satisfactory level of service delivery. The Contractor shall be expected to undertake and successfully implement and achieve these requirements through a process of employing best practice techniques and consultation with FSO/Client and shall be continually strive to improve the level of services.

21.2 Those services include but not limited to:

21.2.1 Operation Services:

- a) Computerised Maintenance Management System (CMMS)
- b) Landscaping
- c) Waste and Redundant Materials

21.2.2 Maintenance Services:

- a) Mechanical Services
- b) Electrical Services
- c) Civil, Structural & Architectural Services

21.3 Contractor shall provide signage to notify on the corrective and preventive maintenance work in progress as accepted by FSO. The signage shall include but not limited to type of work, reference number of work order, execution time and date, contact person in charge and company logo. The Contractor shall use the proper size of signage to make sure it is readable and understandable.

21.4 The Contractor may/shall take digital photo before, during and after any related work done (especially on maintenance services, landscaping and custodial services) as the evidence and for future reference. These digital photos shall be deposited throughout the contract period in a properly labelled folders (e.g. Work Order numbers, date, work trade) to ensure the quick discovery when FSO require it as an evidence of the work done. The photos can be uploaded

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	SECTION E: FACILITY ENGINEERING OPERATION AND MAINTENANCE SERVICES	Section Revision: 2 Page 59 of 85

to CMMS if the system allowed such storage or else in other reliable manner (i.e. hard disk drives, cloud based storage, etc.).

- 21.5 The Contractor shall manage the operation and maintenance services of the buildings, infrastructures, facilities, services and spare parts equipment to Malaysian Standard (MS) or equivalent standard accepted by FSO to ensure the life cycle of all assets are optimized at all times.
- 21.6 The Contractor shall report to FSO the needs to replace major asset/equipment due to obsolescence / Beyond Economical Repair (BER). The Contractor shall submit replacement proposal of equivalent or better specifications/features for FSO approval in accordance to Provisional Quantities or Provisional Sum or Variation Order format. The Contractor shall facilitate the replacement of asset/equipment registration process for the Client. The replacement proposal must be certified by an expert, if necessary.
- 21.7 The Contractor shall provide the Planned Preventive Maintenance (PPM) master list according to the best practices for the operation and maintenance services subject to discussion and acceptance from FSO before the commencement of work. Government Building Scheduled Maintenance Guideline (GBSMG) of the latest version can be used as a reference in developing PPM master list. Government Building Scheduled Maintenance Guideline (GBSMG) contains details on the building services and maintenance requirements for the assets. This guideline is a statement of typical maintenance schedule and frequency for the asset. Further reference to manufacturers' recommendation should be made to supplement the guideline. If the Contractor establishes that the maintenance task detailed is inappropriate or incomplete for an asset, the Contractor shall amend and elaborate accordingly the maintenance task and submit to the FSO for acceptance. The Contractor shall give due consideration to the conditions under which the asset is operating and advice the FSO of any changes to the maintenance frequency. The PPM master list shall be reviewed if necessary throughout the contract period.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	SECTION E: FACILITY ENGINEERING OPERATION AND MAINTENANCE SERVICES	Section Revision: 2 Page 60 of 85

- 21.8 The Contractor shall provide proactive initiatives with regards to availability and functionality of the asset in order to enhance the preventive aspect of maintenance. Such initiative shall be treated as any request or complaint made by the Contractor's employee. Proactive request shall be logged as Self Finding to demarcate it from user/customer request or complaint, in accordance to Table 2 Appendix 7B (Parameters & Data Source for APD).
- 21.9 During the performance of the services the Contractor shall proactively ensure that:
- any maintenance activities shall be undertaken to minimize disruption to user/customer;
 - the relevant Authorities and the FSO shall be informed at least one (1) week before any planned maintenance activities are scheduled to be undertaken that may impact on the amenity of user/customer; and
 - affected user/customer shall be informed at least three (3) working days before any planned maintenance activities are carried out
- 21.10 The Contractor shall rectify / repair any defects identified within agreed Execution Time as specified in Appendix 7B Performance Requirements. For assets that are under warranty or DLP, the FSO shall be immediately notified of the defects for further action.
- 21.11 In the event the execution requires more time, the Contractor shall ensure that the failure is contained, the area is secured, and satisfactory temporary execution is in place within the allotted Execution Time. The prevailing Execution Time shall then be agreed between the Contractor and the FSO.
- 21.12 All maintenance services carried out by the Contractor shall be undertaken by qualified personnel as per item 9.0 Human Resource Requirement. For compliance of Act and specific equipment, (i.e. Electrical, lift, BCS, UPS, CCTV, CMS Link, etc.) the Contractor shall appoint the competent/qualified/ authorized/principal OEM personnel and accepted by FSO to carry out the maintenance works. The service sheet/report shall be included in Monthly Performance Review report to be submitted to FSO together with further action

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	SECTION E: FACILITY ENGINEERING OPERATION AND MAINTENANCE SERVICES	Section Revision: 2 Page 61 of 85

taken by the Contractor if any recommendation for repair needed in the service sheet/report. Failing which, FSO can appoint other party to repair and the cost shall be borne by the Contractor.

21.13 The Contractor shall perform testing & commissioning for operation and maintenance activities as per specification.

21.14 The Contractor shall ensure that all facility management and maintenance in respect of the services is undertaken in a diligent and fully auditable manner.

22.0 OPERATION SERVICES

22.1 Help Desk Services

The Contractor shall set up a Help Desk to manage user/customer requests for information, work/service requests and incident reports.

22.1.1 The Contractor shall provide a central telephone contact number that is reachable and workable at all time and email address.

22.1.2 The Contractor shall receive, manage and record all telephone calls, SMS, email and any messaging and/or application system available for all enquiries and work/service requests at the Help Desk.

22.1.3 The Contractor is required to provide 24 hours, 7 days per week basis the following services as and when necessary and specified by FSO but not limited to:

- a) data entry of enquiries, work/service requests and incident reports;
 - b) data entry of work request progress and completion status including the cost data.
 - c) provision of enquiry reference numbers to callers and the FSO on request;
 - d) tasking of appropriate staff with work orders, maintenance instructions and incident alerts and details;
 - e) provision of progress reports and completion reports to the FSO;
- and

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	SECTION E: FACILITY ENGINEERING OPERATION AND MAINTENANCE SERVICES	Section Revision: 2 Page 62 of 85

f) provision of performance reports to the FSO.

22.1.4 The Contractor shall ensure any telephone call to the Help Desk is answered within three (3) rings.

22.1.5 The Contractor shall ensure that an accurate initial response to any enquiry is provided immediately, and in the case where the information is not readily available, within 30 minutes from the request logged by Help Desk Service.

22.1.6 The Contractor shall prepare and submit to the FSO a Help Desk status report to be included in the Monthly Monitoring report in table form comprising of the following:

- a) report on the number of calls handled;
- b) report on the rates of success in meeting time-frames for enquiries;
- c) report on reasons for any delays or inability to meet response times;
- d) report details of complaints, including complainant's contact information;
- e) report on any trends with respect to the subject matter of enquiries received by the Contractor; and
- f) details report of any suggested or implemented improvements to procedures which should improve efficiency, or actions which could improve value for money.

22.1.7 The Contractor shall continuously share information and make aware to the users in regards to complaint process and procedures.

22.2 CMMS Services

22.2.1 The Contractor shall register and update all data and information in CMMS according to the latest version Technical Specifications CMMS JKR and able to integrate with Government's Asset Management Information System or any accepted application system by the FSO.

22.2.2 The Contractor shall ensure the FSO has real time, online access to system at all times.

22.2.3 The Contractor shall use the system to:

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
SECTION E:	FACILITY ENGINEERING OPERATION AND MAINTENANCE SERVICES	Section Revision: 2
		Page 63 of 85

- a) maintain a current record of the type, class, nature, location and composition of all assets including spare parts (Asset Inventory);
- b) maintain a record of the current, historical and projected future condition and value of each asset including detailed records of the repair or replacement of asset items or asset sub-items to assist in establishing the residual design life, useful life and whole life of the asset items and asset sub-items;
- c) schedule all preventive maintenance and scheduled corrective maintenance activities; and
- d) maintain a record of the nature, extent, quantity, location, time, cost and type of any maintenance works performed by, or program to be performed by the Contractor.

22.2.4 The Contractor shall ensure Asset Inventory data is maintained by the inclusion, removal or revision of data as the Asset Inventory is changed as a result of work undertaken by the Contractor according to the latest version Technical Specifications CMMS JKR.

22.2.5 The Contractor shall use the CMMS to document the regular inspection of asset items and asset sub-items. It shall also document any failure to meet performance requirements. The CMMS shall indicate an appropriate maintenance response and execution time for every work order.

22.2.6 The Contractor shall use an appropriate mix of planned, preventive, program repair and condition-based maintenance strategies particularly developed for each asset item and asset sub-item.

22.2.7 All the above hardware, software, licenses, and data including the list of assets, asset maintenance history, locations of inventory, updated BIM model, updated drawings, and all records captured in the system shall remain the property of the Government. The Contractor shall officially hand over all such information to the FSO every six (6) months throughout the contract period. At the end of the contract, the Contractor shall also hand over the same information to the successor Contractor to ensure continuity in the management, operation, and maintenance of the assets. The Contractor shall ensure that all digital information, including BIM data and CDE contents, is accessible and

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	SECTION E: FACILITY ENGINEERING OPERATION AND MAINTENANCE SERVICES	Section Revision: 2 Page 64 of 85

transferable to the Government in accordance with the latest Government digital asset management standards. The Contractor shall also submit a BIM Asset Information Update Report together with each handover to verify the accuracy and completeness of the updated BIM data.

22.2.8 The system shall have a capability to notify user through email for data updating at certain cut-off date for performance monitoring purpose by the FSO.

22.3 Landscaping

22.3.1 Contractor shall provide landscaping services which comprise softscape and hardscape.

22.3.2 The scope under landscaping services are (but not limited to):

a) Softscape:

- i. Trees
- ii. Shrub
- iii. Ground Cover
- iv. Close Turfing

b) Hardscape:

- i. Water Fountain (electrical and mechanical element to be put in respective discipline)
- ii. Concrete Paver Slab
- iii. Shelter
- iv. Playground Amenities (Swings, See-saw etc.)
- v. Seating and tables

22.3.3 The Contractor shall carry out all services such as cleaning, beautification and up keeping of all plants and operating and maintaining all related structures associated with landscape to ensure the quality of the landscape area.

22.3.4 The Contractor shall replace dead plants with the same or equivalent plants subject to discussion and approval from FSO.

22.3.5 The Contractor shall always use the correct tools, safe and good quality materials recommended by qualified personnel in accordance to standards and regulations.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	FACILITY ENGINEERING OPERATION AND MAINTENANCE SERVICES	Section Revision: 2
		Page 65 of 85
SECTION E:		

22.4 Waste and Redundant Materials

22.4.1 The Contractor shall prepare Waste and Redundant Materials Management Plan (WRMMP) to be incorporated into the FMM Work Program Plan for FSO's acceptance. The Contractor shall submit revised WRMMP whenever required. The Contractor shall comply with the accepted WRMMP.

22.4.2 Waste, redundant materials and rubbish arising from the execution of the services shall be disposed-off from the sites and all areas kept clean and tidy by the Contractor.

22.4.3 The Contractor shall carry out a complete, adequate and proper waste disposal system from the buildings and the surrounding area associated with the building to a waste disposal centre designated in the building compound. The Contractor shall collect all general waste in a manner which will not interfere with the normal functioning of the areas concerned nor cause any nuisance to the user/customer and dispose of from the site at the local authority approved designated area. The method of waste disposal shall be strictly in accordance to the requirements of the Government, local authority and current legislation.

22.4.4 The Contractor shall supply adequate number of environmental friendly bags (including for the waste paper baskets) which shall conform to the requirement of relevant standard. If there is a central storage bin in collecting route, the bin should have hinged closing lids.

22.4.5 The Contractor shall ensure all waste generated on site shall be managed in accordance with the Solid Waste and Public Cleansing Management Act 2007 and Environmental Quality Act 1974.

22.4.6 The Contractor shall provide Roll-On Roll-Off (RORO) for construction waste and Mobile Garbage Bin (MGB) for domestic waste as and when necessary.

22.4.7 The Contractor shall provide a minimum of one (1) location on site for segregation and collection of construction and domestic waste.

22.4.8 The Contractor shall appoint a licensed Contractor(s) to collect the construction waste, scheduled waste and domestic waste from the site to the approved locations for disposal or to recycle the waste.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
SECTION E:	FACILITY ENGINEERING OPERATION AND MAINTENANCE SERVICES	Section Revision: 2
		Page 66 of 85

23.0 MAINTENANCE SERVICES

23.1 Maintenance Schedule and Frequency

23.1.1 The Contractor may follow the maintenance schedule and frequency laid in the latest version of Government Building Scheduled Maintenance Guideline (GBSMG) as a minimum guide to prepare the maintenance schedule and frequency for FSO's acceptance as explained in item 21.7.

23.1.2 Inspection, repair and replacement record shall be verified by the Facility Manager. This report (E.g. Specialist Report, PPM report, Work Order Report, Etc.) shall be submitted to the FSO as a supporting document for the monthly payment together with the invoice.

23.2 Verification of Completed Work / Deduction

23.2.1 All completed repair works for the Asset shall be verified as follows:

- a) Colour photos taken before, during and after repair works as instructed by the FSO. All the photos shall comprise with all the necessary information such as date, location and description/caption using a proper equipment (e.g. small whiteboard to show the information or other method to show the progress work);
- b) Sign-off form by the complainant when applicable.

23.2.2 List of all completed works / closed work orders shall be verified by Facility Manager and shall be closed in the system (CMMS). The hard copies shall be submitted to the FSO's office for evidence clarification during monthly payment process.

23.2.3 All completed works as per maintenance schedules shall be verified by Facility Manager before monthly payment is processed.

23.3 Parts and Consumables Item

- a) All parts and consumables item shall be provided by the Contractor for all related area and deemed to be included in the contract sum unless it is already stated in provisional quantities.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	SECTION E: FACILITY ENGINEERING OPERATION AND MAINTENANCE SERVICES	Section Revision: 2 Page 67 of 85

b) The Contractor shall be required to supply parts and consumables item inclusive but not limited to the following:

- i. Lubricants for bearings and joints as well as contact lubricants for switchgear (lubricants for switchgear shall be silicon type.)
- ii. Electrolytes and distilled water for batteries
- iii. Electric contact points for switchgear, relays and control gears
- iv. Cotton waste, soap detergents and other cleaning substance for cleaning purposes.
- v. Toilet papers and c-fold papers.
- vi. Air freshener for telephones and toilet.
- vii. Lamps and luminaries accessories (unless specified in provisional quantities)
- viii. Coolants.
- ix. Fuel for generator sets and diesel pump (fire fighting)
- x. Engine oil
- xi. Carbon brushes
- xii. Belting
- xiii. Gland packing
- xiv. AHU filters, air-conditioning split unit filters etc (unless specified in provisional quantities)
- xv. Door accessories (unless specified in provisional quantities)
- xvi. Chemicals and detergents for cleaning and water treatment.
- xvii. Fertilizers, pesticides and chemical for softscapes.
- xviii. Any other consumables related to the scope of works in this contract unless specified in provisional quantities.

23.4 Mechanical Services

23.4.1 The Contractor shall put in place and manage professionally, corrective maintenance, proactive maintenance, planned preventive maintenance, root cause failure analysis, testing, inspection and certification regimes, utilization trend analysis, condition based systems and predictive techniques, for its buildings and all installed assets, machineries, equipment, fixtures, fittings, furnishings, finishes and the like, necessary as a minimum to:

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
SECTION E:	FACILITY ENGINEERING OPERATION AND MAINTENANCE SERVICES	Section Revision: 2
		Page 68 of 85

- a) Meet the requirements and recommendations of the manufacturers, including all warranty conditions.
- b) Comply with all legislations, regulations, by-laws and codes of practice as accordance in item 5.0.
- c) Maintain optimum operation, efficiency and availability.
- d) Maintain major assets and equipment availability, subject to planned downtime for maintenance purposes.
- e) Maintain security equipment and systems to achieve best practice standards.
- f) Maintain operations and maintenance manuals, asset registers, maintenance records and statutory documentation.
- g) Provide physical protection for all assets from any nuisance as per FSO instruction and approval.
- h) Ensure the safety of all user/customer and members of the public.
- i) Ensure the business of all user/customer is not unduly interrupted or disturbed.
- j) Ensure the availability of safe and conducive working environments.
- k) Air-conditioned areas shall generally be maintained at 24°C ±1°C Dry Bulb and Relative Humidity within 55% - 70% and other parameters related to MS1525: 2014 or latest edition
- l) Noise and vibration control of the mechanical equipment shall comply and not limited to the following Codes and Authorities of latest edition:
 - i. ASHRAE 2003 (Chapter 47 – Noise and Vibration Control)
 - ii. SMACNA
 - iii. AHRI 885-2008
- m) Ensure all the test equipment must calibrated accordingly yearly by the authorized body.

23.4.2 Mechanical Services inclusive but not limited to the following systems and associated asset;

- a) Air conditioning and Mechanical Ventilation (ACMV) System.
- b) Cooling and refrigeration system.
- c) Pumping systems
- d) Toilet Flushing Systems

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
SECTION E:	FACILITY ENGINEERING OPERATION AND MAINTENANCE SERVICES	Section Revision: 2
		Page 69 of 85

- e) Hot water systems
- f) Cold water plumbing and sanitary
- g) Vertical transportation
- h) Horizontal transportation
- i) Building Management System (BMS)
- j) Gondola and other maintenance equipment (sky lift /scissor lift).
- k) Automated garden watering and fountain system.
- l) Incinerator system
- m) Fire Protection system
- n) Liquefied Petroleum Gas (LPG)/ Compressed Natural Gas (CNG) systems
- o) Kitchen equipment
- p) Swimming pool and Hydro Therapy Pool system
- q) Medical Gases Pipelines System
- r) Any other system related to the scope of works in this contract.

23.4.3 The Contractor shall ensure that all groups of assets or systems and interconnections and ancillary components which may not be referred to in the Specification in general terms are included for maintenance within the contract. It is deemed that the related distribution and interconnection systems items, both internal and external to the sites, are included in the maintenance tasks, for example, including but not limited to:

- a) volume control dampers, motorized dampers, fire and smoke dampers,
- b) pipeline valves, drain cocks, auto and manual air vents,
- c) piping work,
- d) flushing provisions,
- e) ductwork cleaning using latest technology (upon FSO request/provisional item)
- f) temperature and pressure gauges,
- g) insulation, support brackets, anchors, guides,
- h) remote controls and controllers associated with asset and equipment,
- i) controls and controller's integral with asset and equipment,

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
SECTION E:	FACILITY ENGINEERING OPERATION AND MAINTENANCE SERVICES	Section Revision: 2
		Page 70 of 85

- j) condensate lift pumps in cooling units,
- k) water savers,
- l) pressure vessels,
- m) pumps (fire, chilled water, domestic water and sewerage treatment plant pump)
- n) heat exchangers (cooling, domestic water),
- o) pressurization units,
- p) calibration of sensors (control and monitoring),
- q) pressure relief dampers,
- r) terminal air distribution devices,
- s) water storage tanks,
- t) pressure relief and reducing valves,
- u) strainers,
- v) flush valve,
- w) flow rate meters.

23.4.4 The Contractor shall familiarize with the location and operation of the asset. The Contractor shall proceed regularly and diligently with the services while taking all necessary measures to ensure that no delays to the work program. The Contractor shall ensure that he is fully familiar with the operation and maintenance obligations prior to taking responsibility for the asset.

23.4.5 Due to the interconnection of various assets and sensitivity due to inoperability, no asset shall be isolated or disconnected, unless the asset is redundant (equipped with standby unit), without prior approval of the FSO. A method statement shall be required for asset isolation activities.

23.4.6 The Contractor shall carry out the services in a cost effective and energy efficient manner, and shall, as an on-going proactive process, inform the FSO of any steps that may contribute to the cost reduction while maintaining the required level of service and operation.

23.4.7 The Contractor shall note that controls, including but not limited to parameters, set points and times, shall not be adjusted outside the design criteria without the approval of the FSO.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	SECTION E: FACILITY ENGINEERING OPERATION AND MAINTENANCE SERVICES	Section Revision: 2 Page 71 of 85

23.4.8 Where the general instruction terms of check, examine, inspect appear in the maintenance tasks the Contractor is to report and carry out any remedial works as required by the contract or in other situations report abnormalities on the Service Sheet or within the Monthly Performance Review report as appropriate. The Contractor shall report any urgent remedial works without delay to the FSO. The use of digital photography is encouraged to demonstrate the condition of any asset.

23.5 Electrical Services

23.5.1 The Contractor shall put in place and manage professionally, corrective maintenance, proactive maintenance, planned preventive maintenance, root cause failure analysis, testing, inspection and certification regimes, utilizing trend analysis, condition based systems and predictive techniques, for its buildings and all installed assets, machineries, equipment, fixtures, fittings, furnishings, finishes and the like, necessary to achieve performance, asset availability and service availability as a minimum to:

- a) Comply with all legislations, regulations, by-laws and codes of practice as accordance in item 5.0
- b) Meet the requirements and recommendations of the manufacturers, including all warranty conditions.
- c) Maintain optimum operation, efficiency and availability to achieve best practice standards.
- d) Maintain major assets and equipment availability, subject to planned downtime for maintenance purposes.
- e) Maintain operations and maintenance manuals, asset registers, maintenance records and statutory documentation.
- f) Provide physical protection for all assets from any nuisance as per FSO instruction and approval
- g) Ensure the safety of all user/customer and members of the public.
- h) Ensure the business of all user/customer is not unduly interrupted or disturbed.
- i) Ensure the availability of safe and conducive working environments to meet IEC/ IES Standards, JKR Electrical Specification or existing

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
SECTION E:	FACILITY ENGINEERING OPERATION AND MAINTENANCE SERVICES	Section Revision: 2
		Page 72 of 85

design specification or other requirement subject to FSO's instruction.

23.5.2 Electrical Services inclusive but not limited to the following systems and associated asset;

23.5.2.1 High Voltage System

- a) Switchgears,
- b) Transformers,
- c) Cabling System
- d) Other and all accessories

23.5.2.2 Low Voltage System

- a) Low voltage power distribution systems, isolation transformers and switchgears.
- b) General/special lighting luminaires.
- c) Switches (including firemen switch), Switch Socket Outlet and isolators
- d) Busduct/Cabling system
- e) Emergency and Exit Light luminaires
- f) Street and other external lighting luminaires (including street lighting, bollard, garden light, In-ground luminaires, facade light, signal light and others.
- g) Lightning protection system.
- h) Earthing system
- i) Aircraft Warning Lights / Aviation Lights / Obstacle Lights
- j) Instant Water Heater & Hand Dryer

23.5.2.3 Extra Low Voltage (ELV) System

- a) Security System
 - i. Closed Circuit TV (CCTV) System
 - ii. Card Access Management System (CAMS)
 - iii. Visitor Management System (VMS)
 - iv. License Plate Recognition & Face Detection System
 - v. Perimeter Intrusion Detection System (PIDS)
 - vi. Guard Tour System
 - vii. Under Vehicle Security System

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
SECTION E:	FACILITY ENGINEERING OPERATION AND MAINTENANCE SERVICES	Section Revision: 2
		Page 73 of 85

- viii. Boom gate & Bollard System
 - ix. Metal Detection System
 - x. X-ray Baggage Scanner
 - b) Communication System
 - i. Public Address System
 - ii. Conference System
 - iii. Video Conference System
 - iv. Fireman Intercom System
 - v. Intercom (Audio/Video) System
 - c) Information System
 - i. SMATV/FM System
 - ii. Queue System
 - iii. Master Clock System
 - iv. Electronic Display Board System
 - v. Electronic Score Board System
 - vi. Electronic Info Kiosk System
 - d) Audio Visual System
 - i. Sound Reinforcement System,
 - ii. Video Projection System,
 - iii. Integrated Audio Visual System
 - e) Stage System
 - i. Stage Lighting System,
 - ii. Motorized Stage Curtain System

23.5.2.4 Information and Communications Technology (ICT) System

- a) Structured Cabling System
- b) Data Centre
- c) Network Operating System
- d) Servers & Storage
- e) Workstation Equipment (PC, Printers, Scanner, Reader etc.)
- f) Network Switches, Wireless Network System, Controllers etc.
- g) Applications & Licenses
- h) Telecommunication System

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	SECTION E: FACILITY ENGINEERING OPERATION AND MAINTENANCE SERVICES	Section Revision: 2
		Page 74 of 85

23.5.2.5 Emergency Power Supply System

- a) Standby Generator sets
- b) Uninterruptible Power Supply (UPS)

23.5.2.6 Renewable Energy (EE&RE) Equipment

- a) Solar Photovoltaic System

23.5.2.7 Any other system related to the scope of works in this contract.

23.5.3 The Contractor shall conduct electrical inspection and testing and shall comply with regulation and JKR Electrical Inspection and Testing Plan (ITP).

23.5.4 The Contractor shall ensure that all assets or systems and interconnection both internal and external to the sites; and ancillary components which may not be referred to in the Specification in general terms are deemed to be included for maintenance within the contract

23.5.5 The Contractor shall familiarize with the location and operation of the asset. The Contractor shall proceed regularly and diligently with the services while taking all necessary measures to ensure that no delays to the work program. The Contractor shall ensure that he is fully familiar with the operation and maintenance obligations prior to taking responsibility for the asset.

23.5.6 Due to the interconnection of various assets and sensitivity due to inoperability, no asset shall be isolated or disconnected, unless the asset is redundant (equipped with standby unit), without prior approval of the FSO. A method statement shall be submitted prior asset isolation activities.

23.5.7 The Contractor shall carry out the services in an effective and efficient manner, and shall, as an on-going process, inform the FSO of any steps that may contribute to the cost reduction while maintaining the required level of service and operation.

23.5.8 Any asset or system controller's parameter, including but not limited to set points and times, shall not be adjusted outside the design criteria without the approval of the FSO.

23.5.9 Where the general instruction terms of check, examine, inspect appear in the maintenance tasks the Contractor is to report and carry out any

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
SECTION E:	FACILITY ENGINEERING OPERATION AND MAINTENANCE SERVICES	Section Revision: 2
		Page 75 of 85

remedial works as required by the contract or in other situations report abnormalities on the Service Sheet or within the Monthly Performance Review report as appropriate. The Contractor shall report any urgent remedial works without delay to the FSO. The use of digital photography is encouraged to demonstrate the condition of any asset.

23.6 Civil, Structural & Architectural Services

23.6.1 The Contractor shall put in place and manage professionally, corrective maintenance, proactive maintenance, planned preventive maintenance, testing, inspection and certification regimes, utilizing trend analysis, condition based systems and predictive techniques, for the site, its buildings and sites, and all installed asset, machinery, equipment, fixtures, fittings, furnishings, finishes, and the like, necessary as a minimum to:

- a) Meet the requirements and recommendations of the manufacturers, including all warranty conditions,
- b) Comply with all legislations, regulations, by-laws and codes of practice as accordance in item 5.0,
- c) Maintain optimum operation, efficiency and availability,
- d) Maintain operations and maintenance manuals, asset registers, maintenance records and statutory documentation,
- e) Protect Government assets,
- f) Ensure the safety of all user/customer and members of the public,
- g) Ensure the business of the user/customer is not unduly interrupted or disturbed.

23.6.2 The maintenance shall be provided for all areas of the asset that are the responsibility of the FSO and shall include the maintenance of Building Elements, Civil, Structural & Architectural and Public Health Services.

23.6.3 Civil, Structure and Architecture inclusive but not limited to the following:

23.6.3.1 External

- a) Fencings and Gates.
- b) Floors & Flooring.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
SECTION E:	FACILITY ENGINEERING OPERATION AND MAINTENANCE SERVICES	Section Revision: 2
		Page 76 of 85

- c) Walls and wall features
- d) Swimming pools, pools and fountains.
- e) Drainage.
- f) Stairs
- g) Handrails and grilles
- h) Signage (including street directories and advertising billboards).
- i) Roofing
- j) Rain water goods (inclusive of gutter)
- k) Ceiling (inclusive supporting system)
- l) Structural framework.
- m) Roads, Hard Standing, Road/ Parking Line Marking and Road Furniture.
- n) Retention / Detention Pond, Monsoon Drains, Culvert, Sumps, Grating and External Drains.
- o) Footpaths
- p) Expansion Joints
- q) Cycle Paths
- r) External elevated and suction water tank
- s) Sewer Lines, Manholes, Gully Traps
- t) Septic Tank / Sewerage Treatment Plant (STP)
- u) Slopes and Earth Retaining Walls
- v) Water Proofing System
- w) Refuse Bin Centre
- x) Loading Bay
- y) Rain Water Harvesting System

23.6.3.2 Interior

- a) Decoration & Paintwork
- b) Doors
- c) Fixtures and Fittings
- d) Floors & Flooring
- e) Partitions
- f) Walls
- g) Curtain Walls, Windows and Glazing

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	SECTION E: FACILITY ENGINEERING OPERATION AND MAINTENANCE SERVICES	Section Revision: 2 Page 77 of 85

- h) Ceilings (inclusive supporting system)
- i) Staircase
- j) Lift lobby
- k) Carpets
- l) Signage
- m) Structural Framework.
- n) Grease Trap
- o) Sewer Lines, Manholes
- p) Plumbing System
- q) Sanitary System
- r) Cold Water Tank
- s) Water Proofing System
- t) Architectural Energy Efficient Features
- u) Built in Furniture
- v) Handrail and Grille
- w) Blinds and Curtain

23.6.3.3 Structural

- a) Column
- b) Slab
- c) Beams
- d) Roof Trusses

23.6.4 Public Health Services inclusive but not limited to the following systems and associated asset:

- a) Above ground internal waste, soil and rain water drainage. Sewerage system and surface water drainage.
- b) Below ground foul and surface water drainage.
- c) Roadway/roadside drainage
- d) Effluent Tanks & Pumps

23.6.5 Where the general instruction terms of check, examine, inspect appear in the maintenance tasks the Contractor is to report and carry out any remedial works as required by the contract or in other situations report abnormalities on the Service Sheet or within the Monthly Performance Review report as appropriate. The Contractor shall report any urgent

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	SECTION E: FACILITY ENGINEERING OPERATION AND MAINTENANCE SERVICES	Section Revision: 2 Page 78 of 85

remedial works without delay to the FSO. The use of digital photography is encouraged to demonstrate the condition of any asset.

23.6.6 The Contractor shall carry out the cleaning of all drainages regularly and shall undertake monthly program to ensure that all drainages are clear from blockages and free of debris.

23.6.7 The Contractor shall appoint an Engineering Consultant Firm/ Professional Building Surveyor/ Inspector subject to the approval of FSO to conduct a visual checking for building condition once per contract period which includes all main structure integrity. (Roof trusses, main structure bracing the building, shear wall, walls, ceiling, etc.) Any issues regarding building integrity that need further action, the Contractor shall appoint a Consultant/Professional Engineer subject to the approval of FSO to conduct further structural integrity checking. (if required) Scope of work that need to be carried out as follows:

- a) Condition Assessment & Dilapidation Survey
- b) Engineering Factual Report (Endorse by PE)
- c) Site Verification Test (e.g. Coring & Carbonation Test, Ferro scan Test, etc.)

Report shall be submitted to FSO together with suggestion prior completion of the checking.

23.6.8 The Contractor shall provide a corrective building architectural maintenance services on 24 hours per day, 365 days per year basis to address ad hoc emergency, urgent and/ or routine building architectural maintenance requests. The Contractor shall respond to such Service Request within the Response Time to FSO's approval and shall return the affected element(s) to the required Quality Standard within the allotted Execution Time. Corrective building architectural maintenance tasks include, but are not limited to:

- a) Damaged, broken or missing glazing, cladding, tiling, etc.
- b) Damaged or malfunctioning doors ironmongery and fittings.
- c) Damaged wall coverings and painted areas
- d) Other requests received by the help desk (to be determined).

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	SECTION F: FACILITY CUSTODIAL SERVICES	Section Revision: 2 Page 79 of 85

SECTION F: FACILITY CUSTODIAL SERVICES

24.0 HOUSEKEEPING SERVICES

- a) Housekeeping Services provided by the Contractor shall include for all spaces in the building and outside the buildings within the boundary, including both sides external walls and the surrounding area that is associated with the buildings such as roads, driveways, parking areas, footpath, assets, gadgets, fittings, fixtures, equipment, structures and installations situated in the area mentioned which require cleaning services in accordance with the scope and specification to ensure the quality of cleanliness of all office spaces and all external area that is associated with the building except internal area of kitchen, restaurants and cafeterias area which are done by the respective operators themselves.
- b) The Housekeeping Services must include all internal, general and routine cleaning to include floors, carpets, tiles, partition walls, internal walls, suspended ceilings, lighting, furniture and upholstery cleaning, window cleaning, deep cleans of sanitary conveniences and washing facilities, pantries and dining areas, telephones and ICT equipment, consumables and feminine hygiene facilities as well as cleaning of other item that needs periodic cleaning as required.
- c) Cleaning requirements must be determined by stating the method to be deployed using suitable equipment and chemicals accepted by the FSO. It is advisable to consider using Green Product / environmentally friendly chemicals to be used in this contract.
- d) The Contractor shall adequately supply and replenish the relevant replenishment to all the sanitary fittings inside the rest room such as but not limited to hand soap liquid, air fresheners and antiseptics.
- e) The Contractor shall ensure all the equipment, fixtures and finishes are free from dust, smudges, smear, mould, unpleasant odour, stains, streaks and mineral deposits. The Contractor shall also ensure all the equipment is clean and functional. Consumable items shall be in sufficient supply.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
SECTION F:	FACILITY CUSTODIAL SERVICES	Section Revision: 2
		Page 80 of 85

- f) The Contractor shall provide qualified or competent personnel to perform the high-rise building cleaning services. The Contractor shall use the appropriate method (etc. using abseiling or scaffolding or machinery) and equipment need to carry out the high-rise building cleaning services for internal and external of the building. All elements inclusive but not limited to windows, wall, high structures must be clean and clear from dirt, debris and insects. Prior to start of work, all equipment shall be complying with all rules and OSHA regulations. The work shall be done without disruption to the occupants of the building.
- g) If the Housekeeping Services is done by other party appointed by the Government, the Contractor shall collaborate with the other party in order to provide seamless service to user/customer.

24.1 Cleaning Equipment

24.1.1 The Contractor shall ensure proper toilet cleaning maintenance is carried out as specified; the cleaner should have access to appropriate amount of the following list of equipment (not limited to):

- a) Cleaning Tools
- b) trolley and safety signs,
- c) ladder,
- d) floor mop, dust mop and scraper,
- e) bottle spray,
- f) single/ double bucket,
- g) micro fibre cloth,
- h) non-abrasive pad,
- i) toilet brush,
- j) hand gloves,
- k) broom, dustpan and garbage bag,
- l) Protective equipment / clothing,
- m) scrubbing brush and hand brush

24.2 Cleaning Machine

24.2.1 The Contractor should have access to appropriate amount of the following list of machinery (not limited to) in order to ensure proper and satisfactory cleaning services:

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
SECTION F:	FACILITY CUSTODIAL SERVICES	Section Revision: 2
		Page 81 of 85

- a) low speed rotary scrubbing machine with pad holder (brush/ green pad),
- b) high speed machine with 350rpm or higher for buffing works,
- c) dry pick-up vacuum,
- d) water pick-up vacuum,
- e) blower,
- f) high pressure water jet,
- g) hot water extraction

24.3 Chemical Cleaning

24.3.1 The Contractor shall provide proper training to the cleaners in the appropriate use of specific cleaning agents for various types of materials and finishes. Table 3 below can be used as a guide:

Items	Recommended cleaning chemical
Wall / Floor	Neutral based cleaners or disinfectants
Glass / Mirror	Neutral or ammonia based glass cleaner
Sanitary wares	Disinfectants or mild abrasives
Stainless steel / Chrome	Stainless steel or chrome polish
Plastic / PVC	Neutral based cleaner
Toilet and urinal bowls	Disinfectant or mild abrasives

Table 3: Chemical for Cleaning

24.4 Hygienic Services

24.4.1 The Contractor shall adequately supply and replenish the relevant replenishments to all the sanitary fittings inside the rest room such as hand soap liquid, air fresheners, antiseptics and supply sanitary bins (for female toilets).

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
SECTION F:	FACILITY CUSTODIAL SERVICES	Section Revision: 2
		Page 82 of 85

25.0 PEST CONTROL SERVICES

25.1 The Contractor shall provide an effective control and preventive action to all types of pests inside and outside the buildings and the surrounding area that is associated with the buildings to ensure the building and its surrounding is free from pests. The equipment and chemicals used must be safe, good quality and in accordance with the specification relevant standards and current act.

25.2 The Pest Control Services shall provide, manage and operate a comprehensive system of pest control management in accordance with the provisions of this Specification. Types of pests are included but not limited to:

25.2.1 Insects including others but not limited to the following:

- a) Flies;
- b) Cockroaches;
- c) Millipedes;
- d) Centipedes
- e) Wasps;
- f) Stored product insects such as moths;
- g) Lice and mites;
- h) Silverfish;
- i) Ants;
- j) Fleas;
- k) Crickets;
- l) Bees;
- m) Hornets;
- n) Mosquitoes;
- o) Termites;
- p) Borers;
- q) Scorpions.

25.2.2 Rodents and mammals including but not limited to the following:

- a) Rattus Norvegicus (common, brown or Norway rat);
- b) Rattus Rattus (ship or black rat);
- c) Musdomesticus (house mouse);
- d) Grey squirrels;

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
SECTION F:	FACILITY CUSTODIAL SERVICES	Section Revision: 2
		Page 83 of 85

- e) Rabbits;
- f) Foxes;
- g) Feral cats/dogs;
- h) Moles;
- i) Bats;
- j) Monkeys.

25.2.3 Birds including but not limited to the following:

- a) Feral pigeons;
- b) Swallow/ swiftlet;
- c) Sparrow;
- d) Crow

25.2.4 Reptiles including but not limited to the following:

- a) Snakes;
- b) Monitor lizards;
- c) Lizards

25.2.5 Pest Control Services provided are:

- a) Fogging;
- b) Rat baits;
- c) Spraying and baits for termites, ants and cockroaches;
- d) Larvaciding;
- e) Other pest control services that are able to guarantee the safety and health to the occupant and Government assets.

25.3 Statutory/legal compliance

25.3.1 It must be considered in every instance whether the use of any form of chemical (for uses including but not limited to fertilizer, pesticide and herbicide) is strictly necessary before application.

25.3.2 The use of chemicals specifically approved for the purpose for which it is intended must be applied as dictated by the Akta 149: Akta Racun Makhluk Perosak 1974 and the conditions of approval for the chemicals and any other relevant code of practice as issued by the Department of Agriculture (DOA).

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	SECTION F: FACILITY CUSTODIAL SERVICES	Section Revision: 2 Page 84 of 85

25.3.3 All chemicals shall be applied in accordance with manufacturers' recommendation in Material Safety Data Sheet (MSDS) with regards to dilution, application and safety precautions.

25.3.4 The service provider or the person appointed to handle the Pest Control services must be licensed with Lesen Pengendali Makhluk Perosak under Lembaga Racun Makhluk Perosak.

25.4 If the Pest Control Services is done by other party appointed by the Government, the Contractor shall collaborate with the other party in order to provide seamless service to user/customer.

	TECHNICAL SPECIFICATION FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.TS(01).2025
		Date: November 2025
	SECTION G: TECHNICAL ADVICE AND EXPERT PROPOSAL SERVICES	Section Revision: 2 Page 85 of 85

SECTION G: TECHNICAL ADVICE AND EXPERT PROPOSAL SERVICES

26.0 TECHNICAL ADVICE AND EXPERT PROPOSAL SERVICES

- 26.1 The Contractor shall provide proposal report of any finding to improve the FMM Services which are not stipulated in the contract as Technical Advice and Expert Proposal Services at any time throughout the contract or when required by FSO.
- 26.2 If there is any request by FSO, the Technical Advice and Expert Proposal shall be submitted to FSO within the specified time, failing which APD will be imposed.
- 26.3 Upon approval of the proposal report by FSO, a detailed report together with cost for the proposed work shall be prepared and submitted by the Contractor to FSO for approval. The Contractor shall propose at least three (3) option proposals (incorporating the Life Cycle Cost analysis or other practical methodology) for FSO to decide the best option.
- 26.4 The preparation of the detailed reports shall be carried out by qualified personnel as accepted by FSO and the charges/fees of detailed report shall be paid accordingly through appropriate payment means available under FSO jurisdiction.
- 26.5 The Contractor shall ensure the effectiveness of the proposal/reports and shall be responsible for all consequences arises due to failure of the work carried out.

LAMPIRAN 7A1

Maklumat Asas Agensi Pelanggan

a. Kementerian / Jabatan / Agensi Pelanggan:
.....

b. Visi:
.....
.....
.....

c. Misi:
.....
.....
.....

d. Objektif:
.....
.....
.....

e. Jumlah Kakitangan:

Senarai Gred Perjawatan:			
			
			
			

f. Bisnes/Perkhidmatan Teras:

1.
2.
3.
4.
5.

g. Senarai Cawangan/Bahagian: (sertakan Carta Organisasi sekiranya berkaitan)

1.
2.
3.
4.
5.
6.
7.
8.
9.
10.

h. Waktu Beroperasi:

Addendum to Technical Specifications for Building Facility Management & Maintenance

i. Senarai Penyewa Bangunan (sekiranya ada)

1.
2.
3.
4.

j. Kaedah Operasi:

1. Pejabat

☐

2. Orang Awam

☐

i. Kaunter Urusan

ii. Urusan Rasmi (mesyuarat, dsb.)

k. Pelan lantai terkini (Sila sertakan bersama)

l. Keluasan Bangunan :

m. Senarai ruang dan fungsi ruang

No Ruang	Nama Ruang	Fungsi Ruang	Kekerapan Penggunaan Ruang

n. Lain-lain maklumat yang perlu (lampirkan maklumat berkaitan)



JKR COMPUTERISED MAINTENANCE MANAGEMENT SYSTEM (CMMS) SPECIFICATIONS

Version 2.0

Urusetia
Jawatankuasa Spesifikasi CMMS JKR
Bahagian Perundingan Pengurusan Aset
Cawangan Perancangan Aset Bersepadu, Ibu Pejabat JKR Malaysia

Computerised Maintenance Management System (CMMS)

1. The Contractor may use *any* licensed CMMS software which is fulfill the requirement as per item (5) and any *upgrading work as per item (14) for CMMS software shall be borne by the Contractor.*
2. The Contractor shall provide accumulative add on 2 Concurrent Users License throughout the respective contract period.
3. The roles and responsibility of each party involves in this contract are defined as per **attachment A (Matrix Roles and Responsibility for CMMS Software).**
4. The F.S.O. may from time to time to issue further written instruction to comply an accommodate processes; application system; requirement and coding standard as per current instruction issued by the Government of Malaysia. The Contractor shall forthwith comply will all instructions issued to him by the F.S.O.
5. The Contractor must ensure that their proposed *CMMS software* is always compliance, similar or equivalent to these requirements:
 - (a) The system proposed shall provide audit trail capability for tracking of operational & maintenance activities.
 - (b) Helpdesk Function
 - i) Online submission of Work Request with auto number generation and status update.
 - ii) Work request or complaint should be able to deliver by using multiple channels including mobile application, web, email or phone call.
 - iii) Track request and record requester, problem description, and on which equipment and location.
 - iv) Raise work order from the request.
 - v) View request status and /or work order status.
 - vi) Allow attachments on work request.
 - vii) All request to capture requestor's name, Identity Card No, State, District, Agency and facility.
 - viii) Work Request to include escalation flagging through email.
 - (c) Work Order Function
 - i) Able to view and manage all maintenance works and activities.
 - ii) Provide accurate basis for work order costing, analysis, action taken and management of respective facility activities.
 - iii) Audit trail of all activity (material, employee, location, cost, user profile, spares & consumables).
 - iv) Allows prioritizing and categorizing of work order.
 - v) Capture failure code and prevention taken.

- vi) Allows assignment of work order.
- vii) Email alerts capability to inform status of work to requestor if required.
- viii) Allow attachment of external document.
- ix) Able to print Work Order with attachment.

(d) Preventive Maintenance Function

- i) Able to develop and generate schedule works or planned maintenance (work based & scheduling activities) and to be performed on an elapsed time interval.
- ii) Accommodate schedule works based on Time or Actual Completion.
- iii) To be able to schedule works based different scheduling frequencies.
- iv) Able to create a group of schedule work either by location or asset.
- v) Able to upload external documents.
- vi) Allow attachment of external document.
- vii) Able to print with attachment.

(e) Asset Management Function

- i) Asset registration and coding convention should comply to Sistem Kod Aset Tak Alih Kerajaan, *SKATA (Pekeliling Am Bilangan 1 tahun 2018) or the latest version and Garis Panduan Pengumpulan Data dan Pelabelan Aset Tak Alih, PeDATA (Surat Arahan KPKR JKR Bilangan 8 Tahun 2017) or the latest version.*
- ii) Record and manage history on every asset's activity.
- iii) New Asset Requisition Template and quick asset search.
- iv) Capture all assets received and received verification.
- v) Track movement of Immovable & Movable Asset.
- vi) Track all asset warranty and contract.
- vii) Track purchase information and depreciation
- viii) Create and track relationship for Immovable Asset.
- ix) Display work order history for each Asset.
- x) Track bill of material for each Immovable Asset for critical spares management.
- xi) View hierarchy by Immovable Asset tree.
- xii) Able to upload external documents.
- xiii) Allow attachment of external document.

- (f) Asset Premise & Location Function
 - i) Record and manage organization's facility physical locations.
 - ii) Record and manage history on every location on all activities.
 - iii) Location coding structure is to be recorded as per
 - iv) Sistem Kod Aset Tak Alih Kerajaan, *SKATA (Pekeliling Am Bilangan 1 tahun 2018) or the latest version and Garis Panduan Pengumpulan Data dan Pelabelan Aset Tak Alih, PeDATA (Surat Arahan KPKR JKR Bilangan 8 Tahun 2017) or the latest version.*
 - v) Capture location attributes data (user department, occupant, size, and assets of respective location).
 - vi) Able to upload external documents.
 - vii) Allow attachment of external document.
 - viii) Able to print with attachment.
- (g) Material/ Spares Management Function
 - i) Track & manage spares or/and consumables receives, issues and returns.
 - ii) Track adjustment to stock counts due to expiration and obsolescence.
 - iii) Stock count and allotment of spare to work order.
 - iv) Track cost based on standard and average price.
 - v) Issues spares from work order function.
 - vi) Track reorder, min and max level for each spares.
 - vii) Track part balances by multiple sub stores.
 - viii) Track transfer of parts between sub stores.
- (h) Room/ Space Reservation Function
 - i) Log reservation information for rooms and locations.
 - ii) Capture request for rooms/locations reservation – requestor, location, event description, no. of participants, facility required and person in charge.
 - iii) Include report and allow attachments for pictures, free text for notes and suggestions.
- (i) Waste Management Function
 - i) Manage waste collection schedule for all designated location within a facility.
 - ii) Capture planned collection schedule into the system.
 - iii) Generate collection work order for waste collection.
 - iv) Display collection schedule by location for quick view and guide.

- (j) Utility Management Function
 - i) Cater and record utility bills i.e. electricity and water in each facility.
 - ii) Enter and capture meter readings and date of readings.
 - iii) Capture monthly utility bills.
 - iv) Report utility bill trending.

- (k) Post Construction Defect Management Function
 - i) Manage defects during the Defect Liability Period (DLP).
 - ii) Able to view and manage all defects works and activities.
 - iii) Capture remedial action, detail repair information, verification and acceptance.
 - iv) Able to upload external documents.
 - v) Allow attachment of external document.
 - vi) Able to print with attachment.

- (l) Key Performance Indicator Function
 - i) To capture all parameters of KPI in Contracts *as stated in Appendix 7B Performance Requirement for Building Facility Management Maintenance or the latest version* and produce a Monthly Summary Report

- (m) Ascertained Performance Deduction (APD) Function
 - i) To calculate elements of deduction and produce monthly report as specified in this contract, *which refer to the Appendix 7B Performance Requirement for Building Facility Management Maintenance or the latest version.*

- (n) Operation & Management Report Function
 - i) Provide Executive Dashboard.
 - ii) Provide Strategic Management Dashboard and Operational reports for every CMMS functions
 - iii) Operational reports grouped by assets, locations, work orders, work request, preventive maintenance etc.
 - iv) Ability to export to Excel.

- (o) System Administration Function
 - i) Allow configuration of authentication and accessibility levels of users.
 - ii) Assign accessibility and usage in multiple facility and multiple contractors.
 - iii) Able to set accessibility by users by individual function.
 - iv) Set up escalation rules and email designation.

- (p) Training & Support Function
 - i) Provide training conducted by qualified trainer either on-site or at the provider premises.
 - ii) Provide minimum 1 session of training for maximum 10 pax by certified trainer with complete student handouts.
 - iii) Technical support on bugs fixes, patches, and updates of the *CMMS software* through-out the Support period.
 - iv) Support via email, telephone and skype.
 - v) Support based on the contract terms or maximum 5 years whichever comes first.
 - vi) Support service response on a-24 hour business day

- (q) Drawing Management Module Function
 - i) Able to index the drawing documents
 - ii) Allow to upload and store of drawing documents
 - iii) Able to retrieve and view of drawing documents.
 - iv) Able to record drawing history
 - v) Allow attachment of drawing documents format include CAD, JPEG and PDF.

- (r) Cleaning Management Function
 - i) Able to develop and generate cleaning schedule works or planned maintenance (work based & scheduling activities) and to be performed on an elapsed time interval.
 - ii) Able to create Cleaning Inspection List include Inspection Group
 - iii) Accommodate cleaning schedule works based on Time or Actual Completion.
 - iv) To be able to cleaning schedule works based different scheduling frequencies.
 - v) Able to upload external documents.
 - vi) Allow attachment of external document.
 - vii) Able to print with attachment.

6. CMMS software proposed must have security features and high integrity to avoid any possibility of it being manipulated by the user.
7. The contractor must ensure that they will always comply with **Service Level Agreement (SLA) as per attachment B, Jadual Pematuhan Spesifikasi Teknikal as per attachment C and Schedule Payment for CMMS software as per attachment D.**
8. The Contractor is required to install telephone line with stable and adequate permanent internet access (i.e., Unifi) for the purpose of linking to the CMMS proposed. The cost for providing this service shall be borne by the Contractor.
9. The Contractor is responsible to ensure Service Desk personnel are well trained for efficient usage and application of *CMMS Software*.
10. The other costs incurred to make *CMMS Software* fully function shall be borne by the Contractor.
11. All the above hardware, software, license, data, and information captured in the system *should be accessible, readable, and utilizable at any point of time and it shall remain the property of the Government. It shall officially hand over to the F.S.O. in excel and csv format* at the end of the contract to ensure continuity in the management, operation, and maintenance of the assets.
12. Any practice that appears on this site are not clear and have a conflict between CMMS software and Scope of the Contract, shall be brought to the F.S.O. to be discussed and decided upon her.
13. Shall *CMMS Software* go down unexpectedly the Contractor shall run the operation and maintenance activities accordingly in manual approach and urgently to acknowledge the F.S.O.
14. *The contractor must ensure to upgrade their CMMS Software based on the latest requirement from the F.S.O within the contract period as follows:*
 - (a) *Key Performance Indicator (KPI) & Performance Indicators (PI) as per item in Appendix 7B Performance Requirement for Building Facility Management Maintenance or the latest version.*
 - (b) *Preparation of script and transmit the data from the contractor's CMMS software to the Dashboard CMMS JKR. The data files should be thoroughly scanned and validated against an anti-virus scanner with up-to-date virus signatures before being delivered to the Dashboard CMMS JKR.*
 - (c) *Customer Satisfaction Survey Form (Borang Kajian Tahap Kepuasan Pelanggan) JKR.PK(P).11-6 in the Sistem Pengurusan Bersepadu (SPB) JKR or the latest version.*
15. *The contractor shall submit data from CMMS software (i.e., list of assets, asset maintenance history, location of inventory and updated drawings) in excel and cad/rvt/pdf format every six (6) months throughout the contract period for review, comment, and acceptance by the F.S.O.*
16. *The contractor shall ensure to comply with all the requirement of Technical Specifications Compliance Schedule as per attachment C prior to purchasing the CMMS Software and shall be verified by F.S.O.*
17. *The contractor shall anticipate the cost of upgrading, improvement, and data storage of the CMMS Software (Local or Cloud based on MAMPU requirement) throughout the contract period to adapt to the latest technology trends.*

Matrix Roles and Responsibility for CMMS Software

Attachment A

The Contractor	JKR (F.S.O.)
Required to secure the purchase order confirmation with CMMS Provider.	To ensure the Contractor and CMMS Provider has secured the purchase order prior to commence the CMMS implementation and payment is made according to payment terms.
During implementation works: <u>For New Contract</u> • Shall receive the data template from CMMS Provider. • Shall brief the planned timeline to complete the project. • Shall brief the data template to be filled for their respective site. • Shall collect and fill in the data template as required by the data template briefing. • Shall verify and pass the completed soft copy of the data template to CMMS Provider within the stipulated time listed in the planned project timeline. • To answerable any query regarding the standard data template. • Shall issue authorized letter, mentioning that support service will be provided by the CMMS's principal. <u>For Renewal Contract</u> • Shall receive the data template from CMMS Provider. • Shall brief the planned timeline to complete the project. • Shall collect & update new data (if any) and fill in the data template as required by the data template briefing. • Shall brief the status of current and process of collect new data and to fill in the data template to be filled for their respective site. • Shall verify and pass the completed soft copy of the data template to CMMS Provider within the stipulated time listed in the planned project timeline. • To answerable any query regarding the standard data template.	During implementation works: <u>For New Contract</u> • To ensure all data collected and submitted as per JKR's guideline. • To ensure the authorize letter regarding support service provider be submitted accordingly by the contractor. • To verify data submitted to CMMS Provider within the stipulated time listed in project timeline. <u>For Renewal Contract</u> • To ensure all new data collected, updated and submitted as per JKR's guideline. • To verify all new data submitted to CMMS Provider within the stipulated time listed in project timeline.
During data migration works: • Shall provide a soft copy data to CMMS Provider as per data template given • Performing Quality checks on the data collected based on data template. • To validate the converted data • Identifying the required data to be migrated into CMMS Application. • Structuring, codification and porting the verified and finalized data into CMMS Application.	During data migration works: • To ensure validated data is as per Contract.

During Operation & Maintenance Works: • To execute all planned & unplanned work order that has been generated by the system and update the progress status into CMMS. • To ensure application systems to be always functioning.	During Operation & Maintenance Works: • To ensure work order is verified and updated into CMMS before proceeding with payment.
During Training Services: • Shall make available the necessary User Champion who is identified to attend the Training. • Shall identify the course contents, prepare, and develop the functionality training materials including classroom training. • To ensure the training provided will equip for them with knowledge to operate the CMMS. • Shall provide the User Certificate for the valid user only. • <i>Shall provide classroom training and explain on all aspects related to how CMMS functions and operates.</i> • <i>Shall provide the Training Materials</i>	During Training Service: • Shall get involve to the training session in order to get knowledge and understand the usage of CMMS. To evaluate the Contractor on CMMS features and functionalities as part of training exercise and assessment.
Support Services: The Contractor shall cover the following items: a. To ensure CMMS Provider attends to the complaint/request with the approved time. b. Troubleshooting and resolving issues logged. c. The supply of updated / revised releases of Application, inclusive of patches, screens, and enhancements (where applicable) d. Provide advisory services and assist the Contractor regarding on any operational issue on implemented system e. Troubleshooting and fixing issues rose against the standard and customization if applicable. f. Data Recovery. Recovery shall ensure that there is no data loss. Data shall be recovered up to the point of failure g. <i>To submit any problem on the CMMS to F.S.O. and CMMS Provider.</i>	Support Services: • To submit any problem on the CMMS to the Contractor. • To ensure CMMS Provider attends to the complaint/request with the approved time.
During Transition Period: • To notify CMMS Provider the status of contract period ended <i>and to take a necessary action upon getting the notification.</i> • To compile all data and work order management in Work Request, Work Order, Preventive Maintenance, and the relevant modules are being closed and ready to handover to the new Contractor upon getting concern from F.S.O. • To ensure all relevant data of asset and transactional data such as in work request, work order and the relevant modules are being well kept and ready to handover to the new Contractor. • To provide necessary advice and to assist the Contractor regarding on any operational issue on implemented system during transition period.	During Transition Period: • To ensure the Contractor and CMMS Provider being compile for all transactional data and ready to handover to the F.S.O and new Contractor. • <i>To ensure that the Contractor to notify CMMS Provider the contract ended.</i>

SERVICE LEVEL AGREEMENT (SLA)

1.0 SCOPE OF SUPPORT SERVICES

The Contractor shall provide support services to ensure proper, efficient and uninterrupted usage of Application and its Database. The Contractor shall provide of support upon commissioning, as part of its standard package throughout the contract period.

1.1 The Contractor shall provide support services on the Application during the Prime Support Period and Off Prime Support Period as defined below.

a) Prime Support Period

The Prime Support period during office hours is between 7.00 a.m. to 6.30 p.m. excluding any public holiday. Two (2) main support channels during this period include:

- i. Telephone calls
- ii. Email

b) Off Prime Support Period

For the Off Prime Support period i.e. from 6.30pm to 7.00 am including weekends and Public holidays, the following support channels are being made available during this period:

- i. Standby Telephone lines
- ii. Email

1.2 Problem Reporting and Tracking

- a) Reporting of failures or errors on the application or database shall be made through the channels defined in section 1.1 above.
- b) Call Back Response (CBR) time is within 1 hour after the call is logged/incident reported
- c) F.S.O. and the Contractor will determine jointly the severity level of the reported incident
- d) Restoration time is as per the SLA defined in Table 1
- e) The Contractor will monitor and report on the progress of the problem/incident restoration. For severity Class A and B-Critical/Urgent incidents, F.S.O. will be advised at least every other hour as to the status of efforts to resolve the problem/incident. Upon resolution of the problem/incident, the Contractor will seek confirmation from the F.S.O. personnel before closing the case.

1.3 Problem Escalation Procedure

All problems shall be logged via the internal request system collated from the First Line Support Channels and shall be assigned to respective support assistance depending on the severity level of the problem.

1.4 Service Levels and Penalty

SLA shall adhere to the service level defined in Table 1 – Severity Level.

1.5 Remedial Maintenance

- a) The Contractor shall provide remedial maintenance at any time to rectify a breakdown or malfunction of the Application as per the service level defined in Table 1– Severity Level.
- b) The Contractor shall keep the Client informed of the progress in resolving the problem as per the escalation process described.

1.6 Scope of Maintenance

The services shall cover the following items:

- a) Troubleshooting and resolving issues logged by the Client
- b) The supply of updated/revised releases of Application, inclusive of patches, screens and enhancements (where applicable)
- c) Technical advice to improve the efficiency and effectiveness of the Application
- d) Provide advisory services and assist Client regarding on any operational issue on implemented system
- e) Troubleshooting and fixing issues rose against the standard and customization if applicable.
- f) Database back up and compressed exercise.

Table 1: Severity Level

SEVERITY	CODE	DEFINITION	RESPONSE TIME	RESTORATION TIME (including Response Time)
Class A - CRITICAL	(A) Total Business Operations Disruption	Severity Class A is defined as complete application system failure where customer experiencing total loss of connection to the main application. (excluding physical ICT infrastructure)	2 hours on site	Same calendar day
Class B – URGENT	(B) Partial Business Operations Disruption	Severity Class B is defined as a less severe problem whereby connection to the application system can be established. However, operation is highly interrupted by intermittence connection and serious degradation of operational quality. (excluding physical ICT infrastructure)	2 hours on site	Within 24 hours
Class C - MINOR	(C) No critical impact to Business Operations, workaround available	Severity Class C is defined as problem that is not severe or minor lag in application system connection. This will include insignificant minor interruption to the problem which will not have impact on the major operations or the problem is localized on certain group of users with workaround available. (excluding physical ICT infrastructure)	2 hours on site	Next business calendar day

Attachment C

TECHNICAL SPECIFICATIONS COMPLIANCE SCHEDULE

Instructions: Please indicate "Yes "or " No" in the area of compliances in respect of the supply deals. Kindly please state also if the supplier has other specifications that similar or better than the minimum specifications given.

No.	Item	Details	Compliance Yes/No	Other Specifications (please state)
1	Programming	Open Source- PHP		
2	Database	MySQL Version 5.0 and above		
3.	Internet Service Provider (ISP)	Internet Information Service (IIS) or Apache		
4.	Support Browser	Recommended: Internet Explorer, Google Chrome, Mozilla Firefox		

No.	Item	Details	Compliance Yes/No	Other Specifications (please state)
1.	Helpdesk Function	i) Online submission of Work Request with auto number generation and status update.		
		ii) Work request or complaint should be able to deliver by using multiple channels including mobile application, web, email or phone call.		
		iii) Track request and record requester, problem description, and on which equipment and location		
		iv) Raise work order from the requester		
		v) View requester status and / or work order status		
		vi) Allow attachments on work request.		
		vii) All request to capture requestor's name, Identity Card No, State, District, Agency and facility.		
		viii) Work Request to include escalation flagging through email.		
2.	Work Order Function	i) Able to view and manage all maintenance works and activities.		
		ii) Provide accurate basis for work order costing, analysis, action taken and management of respective facility activities.		
		iii) Audit trail of all activity (material, employee, location, cost, user profile, spares & consumables)		
		iv) Allows prioritizing and categorizing of work order.		
		v) Capture Non Compliance code and Comply prevention taken.		

No.	Item	Details	Compliance Yes/No	Other Specifications (please state)
		vi) Allows assignment of work order.		
		vii) Email alerts capability to inform status of work to requestor if required.		
		viii) Allow attachment of external document.		
		ix) Able to print Work Order with attachment.		
3.	Preventive Maintenance Function	i) Able to develop and generate schedule works or planned maintenance (work based & scheduling activities) and to be performed on an elapsed time interval.		
		ii) Accommodate schedule works based on Time or Actual Completion.		
		iii) To be able to schedule works based different scheduling frequencies.		
		iv) Able to create a group of schedule work either by location or asset.		
		v) Able to upload external documents.		
		vi) Allow attachment of external document.		
		vii) Able to print with attachment.		

No.	Item	Details	Compliance Yes/No	Other Specifications (please state)
4.	Asset Management Function	i) Asset registration and coding convention should comply to Sistem Kod Aset Tak Alih Kerajaan, SKATA (Pekeliling Am <i>Bilangan 1 tahun 2018</i>) or the latest version and <i>Garis Panduan Pengumpulan Data dan Pelabelan Aset Tak Alih, PeDATA (Surat Arahan KPKR JKR Bilangan 8 Tahun 2017)</i> or the latest version.		
		ii) Record and manage history on every assets activity		
		iii) New Asset Requisition Template and quick asset search.		
		iv) Capture all assets received and received verification.		
		v) Track movement of Immovable & Movable Asset.		
		vi) Track all asset warranty and contract.		
		vii) Track purchase information and depreciation		
		viii) Create and track relationship for Immovable Asset.		
		ix) Display work order history for each Asset.		
		x) Track bill of material for each Immovable Asset for critical spares management.		
		xi) View hierarchy by Immovable Asset tree.		
		xii) Able to upload external documents.		
		xiii) Allow attachment of external document.		

No.	Item	Details	Compliance Yes/No	Other Specifications (please state)
5.	Asset Premise & Location Function	i) Record and manage organization's facility physical locations		
		ii) Record and manage history on every location on all activities.		
		iii) Location coding structure is to be recorded as per Sistem Kod Aset Tak Alih Kerajaan, <i>SKATA (Pekeliling Am Bilangan 1 tahun 2018) or the latest version and Garis Panduan Pengumpulan Data dan Pelabelan Aset Tak Alih, PeDATA (Surat Arahan KPKR JKR Bilangan 8 Tahun 2017) or the latest version.</i>		
		iv) Capture location attributes data (user department, occupant, size and assets of respective location)		
		v) Able to upload external documents.		
		vi) Allow attachment of external document.		
		vii) Able to print with attachment.		
6.	Material/ Spares Management Function	i) Track & manage spares or/and consumables receives, issues and returns.		
		ii) Track adjustment to stock counts due to expiration and obsolescence.		
		iii) Stock count and allotment of spare to work order.		
		iv) Track cost based on standard and average price.		
		v) Issues spares from work order function.		
		vi) Track reorder, min and max level for each spares.		
		vii) Track part balances by multiple sub stores.		

No.	Item	Details	Compliance Yes/No	Other Specifications (please state)
		viii) Track transfer of parts between sub stores.		
7.	Room/ Space Reservation Function	i) Log reservation information for rooms and locations.		
		ii) Capture request for rooms/locations reservation – requestor, location, and event description, no. of participants, facility required and person in charge.		
		iii) Include report and allow attachments for pictures, free text for notes and suggestions.		
8.	Waste Management Function	i) Manage waste collection schedule for all designated location within a facility.		
		ii) Capture planned collection schedule into the system.		
		iii) Generate collection work order for waste collection.		
		iv) Display collection schedule by location for quick view and guide.		
9.	Utility Management Function	i) Cater and record utility bills i.e. electricity and water in each facility.		
		ii) Enter and capture meter readings and date of readings.		
		iii) Capture monthly utility bills.		
		iv) Report utility bill trending.		
10.	Post Construction Defect Management Function	i) Manage defects during the Defect Liability Period (DLP).		
		ii) Able to view and manage all defects works and activities.		
		iii) Capture remedial action, detail repair information, verification		

No.	Item	Details	Compliance Yes/No	Other Specifications (please state)
		and acceptance.		
		iv) Able to upload external documents.		
		v) Allow attachment of external document.		
		vi) Able to print with attachment.		
11.	Key Performance Indicator (KPI) Function	i) To capture all parameters of KPI in FM Contracts <i>as stated in Appendix 7B Performance Requirement for Building Facility Management Maintenance or the latest version of the said appendix</i> and produce a Monthly Summary Report		
12.	Ascertained Performance Deduction (APD) Function	i) To calculate and produce monthly report on deduction computation based on elements of deduction specified in this contract, <i>which refer to the Appendix 7B Performance Requirement for Building Facility Management Maintenance or the latest version of the said appendix</i>		
13.	Operation & Management Report Function	i) Provide Executive Dashboard.		
		ii) Provide Strategic Management Dashboard and Operational reports for every CMMS functions.		
		iii) Operational reports grouped by assets, locations, work orders, work request, preventive maintenance etc.		
		iv) Ability to export to Excel		
14.	System Administration Function	i) Allow configuration of authentication and accessibility levels of users.		
		ii) Assign accessibility and usage in multiple facility and multiple contractors.		

No.	Item	Details	Compliance Yes/No	Other Specifications (please state)
		iii) Able to set accessibility by users by individual function.		
		iv) Set up escalation rules and email designation.		
15.	Drawing Management Module Function	i) Able to index the drawing documents.		
		ii) Allow to upload and store of drawing documents		
		iii) Able to retrieve and view of drawing documents		
		iv) Able to record drawing history		
		v) Allow attachment of drawing documents format include CAD, JPEG and PDF.		
16.	Cleaning Function	i) Able to develop and generate cleaning schedule works or planned maintenance (work based & scheduling activities) and to be performed on an elapsed time interval.		
		ii) Able to create Cleaning Inspection List include Inspection Group		
		iii) Accommodate cleaning schedule works based on Time or Actual Completion.		
		iv) To be able to cleaning schedule works based different scheduling frequencies.		
		v) Able to upload external documents.		
		vi) Allow attachment of external document.		
		vii) Able to print with attachment.		

**PROGRESS OF PAYMENT SCHEDULE FOR
CMMS SOFTWARE**

For New Site			
Bil.	Activities	% Payment	Delivery Evidence
1	Software Package To configure <i>CMMS Software</i> c/w 2 user licenses to fulfil the requirement of this contract.	20	Site Configuration Acknowledgement by the F.S.O.
2	Data Collection To conduct site survey, requirement study, site data collection/ verification & compilation of data as per Government's Asset Registration Template.	20	Asset Registration Templates Verified by the F.S.O.
3	Implementation Services (Site survey, requirement study, compilation of data collection, data porting, installation & commissioning)	20	Letter of Commissioning from <i>CMMS Software Provider</i>
4	Training Services To conduct end-user and system administration training to ensure the <i>CMMS</i> software will be running accordingly to fulfil the requirement of this contract.	15	3 Certificates from <i>CMMS Provider</i> as Per Below: i) List of Attendances ii) 2 User licenses iii) Support Service
5	Support Service by Year(s) To provide helpdesk via email, phone, etc., technical support including attending meeting (if required), software updates and patches throughout the contract.	25	Support Service Trail (Report) - Monthly/Yearly Service Report

For Renewal Site			
Bil.	Activities	% payment	Delivery Evidence
1	Data Collection To conduct site survey, requirement study, site data collection/ verification & compilation of data as per Government's Asset Registration Template.	30	Asset Registration Templates Verified by the F.S.O.
2	Implementation Services (Site survey, requirement study, compilation of data collection, data porting, installation & commissioning)	30	Letter of Commissioning from CMMS Provider
3	Training Services To conduct end-user and system administration training to ensure the CMMS software will be running accordingly to fulfil the requirement of this contract.	15	3 Certificates from CMMS Provider as Per Below: i) List of Attendances ii) 2 User licenses iii) Support Service
4	Support Service by Year(s) To provide helpdesk via email, phone, etc., technical support including attending meeting (if required), software updates and patches throughout the contract.	25	Support Service Trail (Report) - Monthly/Yearly Service Report

Urusetia
Jawatankuasa Spesifikasi CMMS JKR
Bahagian Perundingan Pengurusan Aset
Cawangan Perancangan Aset Bersepadu, Ibu
Pejabat JKR Malaysia

Emel: bppa@jkr.gov.my

**PERFORMANCE REQUIREMENT
FOR BUILDING FACILITY
MANAGEMENT & MAINTENANCE**

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	PERFORMANCE REQUIREMENT FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.PR(01).2025
		Date : November 2025
	TABLE OF CONTENTS	Section Revision: 2
		Page 1 of 21

DEFINITION	2
SECTION A: PERFORMANCE REQUIREMENTS FRAMEWORK	5
1.1 Purpose.....	5
1.2 Introduction	5
SECTION B: KEY PERFORMANCE INDICATOR (KPI) & PERFORMANCE INDICATORS (PI).....	6
2.1 Key Performance Indicator (KPI) & Performance Indicators (PI).....	6
SECTION C: ASCERTAINED PERFORMANCE DEDUCTION (APD)	8
3.1 APD Evaluation Procedure.....	8
3.2 APD Parameter Establishment.....	11
3.2.1 Service Delivery related to Core Business	12
3.2.2 Asset Performance	16
3.2.3 Building Energy Efficiency.....	18
3.2.4 Safety & Statutory Compliance	18
3.3 APD Calculation	19
3.4 APD Calculation Samples	20
SECTION D: DEMERIT POINTS & WARNING LETTERS	21
4.1 Demerit Points Imposition	21

	PERFORMANCE REQUIREMENT FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.PR(01).2025
		Date : November 2025
	DEFINITION	Section Revision: 2
		Page 2 of 21

DEFINITION

Asset	All assets to enable the Contractor to provide FMM Services in accordance with this technical specification, including but not limited to: -any land or buildings and infrastructure; 1. plant and machineries; 2. equipment and system (together with any warranties in respect of assets being transferred); 3. drawings and technical data; 4. intellectual property rights as attached to assets; 5. any books, reports and records which including all operating and maintenance manuals; and 6. any spare parts, tool and other assets (together with any warranties in respect of assets being transferred).
Ascertained Performance Deduction (APD)	A monetary value implied from calculations of the Contractor's non-performance in achieving the KPI target. The APD value shall be deducted from the Monthly Payments.
APD Max	Maximum Upvalue that can be imposed to the Contractor that is 5%* from MPV.
Client	Asset owner or user/customer's representative who can make decision regarding the Asset & Facility.
Complaint	Any written or verbal notice or report by the Customers/Users/FSO/ FSO's representative given through telephone/fax/electronic media (or any other means agreed by FSO) about not meeting agreed standard of service provided by the Contractor as per scope and technical specification (after verified by the Contractor).
Corrective Maintenance	Work / repair / servicing / testing carried out after failure/ breakdown has occurred and is intended to restore an item to a state in which it can perform to its required function as per designed.

	PERFORMANCE REQUIREMENT FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.PR(01).2025
		Date : November 2025
SECTION B:	KEY PERFORMANCE INDICATOR (KPI)	Section Revision: 2
		Page 3 of 21

Custodial Service	General cleaning/housekeeping, pest& domestic animal control and hygiene services.
Demerit Point	Points imposed when the Contractor fails to meet the Performance Indicator (PI) as implication when the Contractor fails to meet the KPI to counter the inconsistent performance.
Evaluation Month	Agreed period and cut-off date between FSO and the Contractor of the FMM Services evaluation for Monthly Payment purpose
Facility	Collection of Assets which is built, installed or established to serve an entity's needs.
Facility Management and Maintenance (FMM)	Managing, administrating, supervising all aspect of asset including but not limited to facility management services, facility engineering operation and maintenance services, and consultation services within the asset's boundaries.
Facility Risk Management	A structured approach to managing effect of uncertainty related to threat, a sequence of human activities including risk assessment, strategies development to manage it, and mitigation of risk using managerial resources.
Facility Superintending Officer (FSO)	Facility Superintending Officer and his successor in office as appointed by the Government under Clause 23 in PWD Facilities Management & Maintenance Form (2016) to act on behalf of the Government for matters stipulated in this Contract.
FSO's Representative (FSOR)	any person appointed by the FSO pursuant to Clause 24 in PWD Facilities Management & Maintenance Form (2016) to perform any duties and exercise any of the rights of the FSO.
Key Performance Indicator (KPI)	Important areas in which the high performance is needed to ensure the success of this contract
Monthly Payment Value (MPV)	The value established from Monthly Payment Plan (MPP) and the actual work done for the Evaluation Month as the basis for APD calculation. MPV shall exclude Preliminaries, Management & Supervisory Staff Remuneration, Provisional Sum and Provisional Quantity items.
Pending / Backlog Work Order	Work Order generated in the previous Evaluation Month which is still open

	PERFORMANCE REQUIREMENT FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.PR(01).2025
		Date : November 2025
SECTION B:	KEY PERFORMANCE INDICATOR (KPI)	Section Revision: 2
		Page 4 of 21

Performance Indicator (PI)	A measurement tools to indicate the performance of the actions that contribute towards KPI
Preventive Maintenance	A work / repair / servicing and testing carried out at pre-determined intervals period to prescribed criteria, intended to reduce failure, environmental damage or the performance degradation of an item to prolong the life expectancy of the equipment.
Planned Preventive Maintenance (PPM)	A planned and scheduled work to carry out the preventive maintenance. The Planned Preventive Maintenance also includes Routine Inspection and Predictive Maintenance / Condition Based Maintenance.
Routine Inspection	Daily/weekly activities of check/inspect item form Planned Preventive Maintenance schedule in Facility Engineering Operation & Maintenance Services and also in Facility Custodial Services.
Schedule Corrective Maintenance (SCM)	Scheduled Corrective Maintenance (SCM) is a planned maintenance activity scheduled to rectify deficiencies or failures identified through PPM, routine inspections, or user complaints. The activities may involve repair, restoration, or replacement of equipment, and are carried out according to required timelines, availability of spare parts, or during routine maintenance shutdowns.
Self-Finding Work Order	Work Order open by the Contractor's staff resulting from Routine Inspection/other activities
the Contractor	refers to FMM Contractor appointed by the Government to implement the FMM Contract.
Users / Customers	All staffs, tenants, visitors and / or occupiers who will reside, operate and / or have their place of business or work at the asset sites. Users / Customers shall include but not limited to Government agencies, bodies and / or authorities.
Walkthrough Audit	A random audit by FSO/FSOR and/or clients together with or without the Contractor to check the condition of premise, assets and equipment and to cross check with maintenance checklist records whether maintenance work has been done properly and accordingly.

	PERFORMANCE REQUIREMENT FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.PR(01).2025
		Date : November 2025
SECTION B:	KEY PERFORMANCE INDICATOR (KPI)	Section Revision: 2
		Page 5 of 21

SECTION A: PERFORMANCE REQUIREMENTS FRAMEWORK

1.1 Purpose

This document is designed to be the standard Performance Requirement for normal operations of an office building premise. For premises with different functionality and/or specific requirements, the Performance Requirements shall be adjusted accordingly to suit the function and objective of the premise. The main objectives of this document are:

- a) to make sure that the Contractor performs their obligations according to the Contract
- b) to assist FSO& Client in measuring and indicating the Contractor's overall and monthly performance in relation with the specified Key Performance Indicator (KPI)
- c) to assist FSO in quantifying the Contractor's performance into monetary value
- d) to make the deduction value as a reminder/push factor for the Contractor to perform better in delivering FMM Services and at the same time not hindering the Contractor to perform routine operational task due to financial constraint from the deduction to serve as an indicator for triggering the termination process.

1.2 Introduction

- a) This framework is intended to imply a fair payment deduction mechanism for the FMM Services that have not met the specified performance requirement.
- b) This deduction mechanism shall not be applicable for works/services outside the scope of FMM Services.
- c) Issues, requests, complaints, shutdowns or unavailability of services resulted during the implementation of Planned Preventive Maintenance (PPM) Schedule or Scheduled Corrective Maintenance (SCM) shall not be subjected to any deductions.
- d) The performance requirements are governed by a set of Key Performance Indicator (KPI) in Table 1 Section B.
- e) If the Contractor fails to meet the target specified in the KPI, two (2) implications will take effect that are:
 - i. Deduction from monies due to the Contractor through Ascertained Performance Deduction (APD) calculation formula specified in Section C.
 - ii. Imposing Demerit Point according to Table 1 in Section B and triggering the process for termination

	PERFORMANCE REQUIREMENT FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.PR(01).2025
		Date : November 2025
SECTION B:	KEY PERFORMANCE INDICATOR (KPI)	Section Revision: 2
		Page 6 of 21

SECTION B: KEY PERFORMANCE INDICATOR (KPI)&PERFORMANCE INDICATORS (PI)

2.1 Key Performance Indicator (KPI) & Performance Indicators (PI)

Key Performance Indicator (KPI) are the important outcome from this FMM Services that have been derived from the Scope and Technical Specifications. KPI in this contract has been developed in a manner that it will focus on balancing the Customer's perspective, asset performance and Service Provider's ability to deliver the agreed service outcome. The KPI are as follow:

No Key Performance Indicator (KPI)

- i. FMM Service Delivery related to Core Business
- ii. Asset Performance
- iii. Building Energy Efficiency
- iv. Safety & Statutory Compliance

For every KPI, there are Performance Indicators (PI) with their respective targets being specified for the Contractor to achieve. Accomplishments of these PI targets are used as a tool to measure the Contractor's performance in delivering FMM Services for this Contract. These PI have respective weightage that can be adjusted (together with client) according to the criticality of the services.

In the event of the Contractor fails to meet the PI target, deduction of monies through Ascertained Performance Deduction (APD) mechanism will take effect. APD is the monetary value deducted from the Monthly Payment Value (MPV) that should indicate the monthly performance of the Contractor. The mechanism of APD shall be discussed further in Section C of this document.

Another implication when the Contractor fails to meet the KPI is the imposition of demerit points. Demerit points are introduced to counter the inconsistent performance as they are accumulative indicators to show the overall performance from the Contractor throughout the contract. This will be explained in Section D of this document. The imposition of Demerit points shall be one of the triggering point for termination of the Contractor as per Garis Panduan Penamatan Pengambilan Kerja Kontraktor bagi Kontrak Pengurusan Fasilitas dan Penyenggaraan.

	PERFORMANCE REQUIREMENT FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.PR(01).2025
		Date : November 2025
SECTION B:	KEY PERFORMANCE INDICATOR (KPI)	Section Revision: 2
		Page 7 of 21

The Table 1 below shows the list of PI and the relations with KPI

KPI No.	Key Performance Indicators * (KPI)	PI No.	Performance Indicators * (PI)	Weightage * (%)	Demerit Point *
1	FMM Service Delivery related to Core Business	1A	Customer Satisfaction Survey rating $\geq$ 80%	5	5
		1B	Customer Rating in Corrective Work Order sheet $\geq$ 70%	5	5
		1C	Response Time 100% met	5	5
		1D	Execution Time $\geq$ 95% (refer schedule A)	5	5
		1E	Pending/Backlog Work Order Completion 100% (refer schedule B)	5	5
		1F	Self-Finding Work Order $\geq$ 80% from total Work Order	10	10
		1G	Cleaning Performance $\geq$ 85%**	6	6
		1H	Pest Control Performance $\geq$ 95%**	5	5
		1I	Critical Services $\geq$ 95% available	8	8
		1J	Normal Services $\geq$ 85% available	7	7
2	Asset Performance	2A	PPM for Architecture, C&S and Landscape*** assets 100% implemented	4	4
		2B	PPM for Mechanical assets 100% implemented	4	4
		2C	PPM for Electrical assets 100% implemented	4	4
		2D	Engineering Reports & Recommendation action 100% taken	4	4
		2E	Work-done as specification and asset quality meet standards	4	4
3	Building Energy Efficiency	3A	Energy/water Conservation programs 100% implemented	4	4
		3B	Building Energy Intensity (BEI) target 100% met (target to be set after energy audit/based on consumption of 12 months before)	3	3
		3C	Utility consumption 100% No wastage	3	3
4	Safety & Statutory Compliance	4A	Relevant Acts & Regulations 100% Comply	3	3
		4B	HSE programs 100% implemented	3	3
		4C	Reports submitted 100% on time with sufficient content	3	3
			TOTAL	100	100

Table 1 Key Performance Indicator (KPI)

	PERFORMANCE REQUIREMENT FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.PR(01).2025
		Date : November 2025
SECTION B:	KEY PERFORMANCE INDICATOR (KPI)	Section Revision: 2
		Page 8 of 21

*Target Percentage, Weightage Value and Demerit Points can be adjusted and agreed by the client during tender stage

** Delete if not applicable

***Delete if Landscape is not applicable in the contract

	PERFORMANCE REQUIREMENT FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.PR(01).2025
		Date : November 2025
SECTION C:	ASCERTAINED PERFORMANCE DEDUCTION (APD)	Section Revision: 2
		Page 9 of 21

SECTION C: ASCERTAINED PERFORMANCE DEDUCTION (APD)

3.1 APD Evaluation Procedure

- FSO/FSOR shall establish the total parameters relevant to the KPI from the data source as in the Table 2.
- FSO/FSOR shall then establish the non-compliance parameters to the KPI from the verification of data sources. These non-compliance parameters are subject to deduction in the APD evaluation.
- All parties involved (FSO, Client, FMM Contractor) shall agree to select a date as the cut-off evaluation date for every Evaluation Month.
- The Contractor shall officially write to FSO/FSOR prior to cut-off date, the justification of the relevant Performance Indicator (PI) that did not achieve the target for the FSO/FSOR jurisdiction.
- For parameters established through random inspections or walkthrough audits, the appropriate sampling quantity shall be determined based on the existing population and must be approved by the FSO. One of the methods to determine the population is illustrated in Figure 1 below:

Table for Determining Sample Size of a Known Population									
N	S	N	S	N	S	N	S	N	S
10	10	100	80	280	162	800	260	2800	338
15	14	110	86	290	165	850	265	3000	341
20	19	120	92	300	169	900	269	3500	346
25	24	130	97	320	175	950	274	4000	351
30	28	140	103	340	181	1000	278	4500	354
35	32	150	108	360	186	1100	285	5000	357
40	36	160	113	380	191	1200	291	6000	361
45	40	170	118	400	196	1300	297	7000	364
50	44	180	123	420	201	1400	302	8000	367
55	48	190	127	440	205	1500	306	9000	368
60	52	200	132	460	210	1600	310	10000	370
65	56	210	136	480	214	1700	313	15000	375
70	59	220	140	500	217	1800	317	20000	377
75	63	230	144	550	226	1900	320	30000	379
80	66	240	148	600	234	2000	322	40000	380
85	70	250	152	650	242	2200	327	50000	381
90	73	260	155	700	248	2400	331	75000	382
95	76	270	159	750	254	2600	335	100000	384
Note: N is Population Size; S is Sample Size					Source: Krejcie & Morgan, 1970				

Figure 1 – Table for Determining Sample Size from a given Population (Krejcie & Morgan, 1970)

	PERFORMANCE REQUIREMENT FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.PR(01).2025
		Date : November 2025
SECTION C:	ASCERTAINED PERFORMANCE DEDUCTION (APD)	Section Revision: 2
		Page 10 of 21

- f) Final APD amount shall be establish based on mutual agreement of both the Contractor and FSO either through a formal meeting or FSO/FSOR's notification letter/email.
- g) Final APD amount imposed for that Evaluation Month shall be sign-off by both parties. If the Contractor is not agree to sign-off within fourteen (14) days after the meeting, the value of APD shall deemed to be agreed by the Contractor as per notification from FSO/FSOR.

Table 2 Parameters & Data Source for APD

KPI No.	KPI	PI No.	PI	Parameters	Data Source
1	Service Delivery related to Core Business	1A	Customer Satisfaction Survey rating $\geq 80\%$	Average score rating percentage (Total score / total respondent)	Quarterly Customer Satisfaction Survey
		1B	Customer Rating in Corrective Work Order sheet $\geq 70\%$	Average score rating percentage (Total score / total respondent)	Customers' rating remarks in Work Order sheet
		1C	Response Time 100% met	Numbers of Corrective Work Order (for the Evaluation Month)	CMMS records/Manual form Request Response sheet/agreed response verification methods
		1D	Execution Time $\geq 95\%$ (refer schedule A)	Numbers of Corrective Work Order (for the Evaluation Month)	CMMS records/Manual form Work Order sheet/agreed completion verification methods
		1E	Pending/Backlog Work Order Completion 100% (refer schedule B)	Numbers of Pending/Backlog Work Order	CMMS records/Manual form Work Order sheet/other records
		1F	Self-Finding Work Order $\geq 80\%$ from total Work Order	Numbers of Corrective Work Order for the Evaluation Month	CMMS records/Manual form Work Order sheet/agreed completion verification methods
		1G	Cleaning Performance $\geq 85\%$	Average marks (percentage) from walkthrough audit done for the Evaluation Month	Walkthrough audit by FSO/FSOR team monthly sampling of Cleaning/Housekeeping Work Order sheet compare to site condition).
		1H	Pest Control Performance $\geq 95\%$	Numbers of completed Pest Control Work Order from walkthrough audit done for the Evaluation Month	Walkthrough audit by FSO/FSOR team monthly sampling of Pest Control Work Order sheet compare to site condition).
		1I	Critical Services $\geq 95\%$ available	Numbers of DAK Ruang affected by downtime of Critical Services / Total DAK Ruang	CMMS records/Manual form Work Order sheet/agreed completion verification methods
		1J	Normal Services $\geq 85\%$ available	Numbers of DAK Ruang affected by downtime of Critical Services / Total DAK Ruang	

	PERFORMANCE REQUIREMENT FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.PR(01).2025
		Date : November 2025
SECTION C:	ASCERTAINED PERFORMANCE DEDUCTION (APD)	Section Revision: 2
		Page 11 of 21

2	Asset Performance	2A	PPM for Architecture, C&S and Landscape assets 100% implemented	Numbers of NCR issued for the Evaluation Month	i. Non-Compliance Report (NCR) issued by FSO/FSOR team through walkthrough audit / Condition Inspection (monthly sampling of PPM /CMMS records compare to site condition). ii. For Engineering Reports & Recommendation, every report for the Evaluation Month needs to be checked. iii. For any work which not follow the specification/asset condition not meet to the quality standards
		2B	PPM for Mechanical assets 100% implemented	Numbers of NCR issued for the Evaluation Month	
		2C	PPM for Electrical assets 100% implemented	Numbers of NCR issued for the Evaluation Month	
		2D	Engineering Reports & Recommendation action 100% taken	Numbers of Report for the Evaluation Month	
		2E	Work-done as specification/ asset quality meet standards	Numbers of NCR issued for the Evaluation Month	
3	Building Efficiency	3A	Energy/water Conservation programs 100% implemented	Numbers of Energy/water Conservation Programs Implemented (for the Evaluation Month)	Actual implemented Energy/water Conservation program
		3B	Building Energy Intensity (BEI) target 100% met (target to be set after energy audit/Based on consumption of 12months before)	Achieve/Not achieve targeted BEI	Actual Building Energy Intensity (BEI) for the Evaluation Month
		3C	Utility consumption 100% No wastage	Numbers of CMMS record/audit finding	CMMS records/walkthrough audit no unnecessary utility usage (lighting, water, A/C)
4	Safety & Statutory Compliance	4A	Relevant Acts & Regulations 100% Comply	Numbers of related Relevant Act/Regulations	Laws, Acts, Standards, Regulations, BOMBA, DOSH
		4B	HSE programs 100% implemented	Numbers of HSE programs implemented for the Evaluation Month	
		4C	Reports submitted 100% on time with sufficient content	Numbers of Reports supposed to submit for the Evaluation Month (refer to Table 2 in Lampiran 7A Technical Specification)	

	PERFORMANCE REQUIREMENT FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.PR(01).2025
		Date : November 2025
SECTION C:	ASCERTAINED PERFORMANCE DEDUCTION (APD)	Section Revision: 2
		Page 12 of 21

3.2 APD Parameter Establishment

3.2.1 Service Delivery related to Core Business

3.2.1A Customer Satisfaction Survey (CSS)

- i. The Contractor shall conduct CSS according to item 14.0 in Appendix 7A Technical Specification for Building Facility Management & Maintenance.
- ii. Questionnaires and methodology of CSS shall be proposed by the Contractor subject to FSO & Client's endorsement.
- iii. The average score rating results (percentage) from CSS marks conducted monthly shall be used to evaluate the PI achievement. APD shall be imposed if the PI condition is not met.

3.2.1B Customer Rating (CR)

- i. Every Corrective Work Order sheet shall provide Customer Rating and Comments section for every user/customer to evaluate, give rating for the work done and comment on the Contractor's performance in handling and resolving complaints.
- ii. The average score rating results (percentage) from CR marks shall be used to evaluate the PI achievement monthly. APD shall be imposed if the PI condition is not met.

3.2.1C Response Time

- i. Execution Time can be traced from CMMS records, manual Work Request response forms/sheet or other agreed response verification records available.
- ii. Response Time Service Level for all Request/Complaint is set within 30 minutes (except for Lift entrapment case within 5 minutes) from the Request/Complaint logged by Help Desk Service.
- iii. Percentage for numbers of Work Order that did not comply with the allocated response time divided by total Work Order for the Evaluation Month shall be used to evaluate the PI achievement monthly. APD shall be imposed if the PI condition is not met.

3.2.1D Execution Time

- i. Execution Time can be traced from CMMS records, manual Work Order forms/sheet or other agreed complete verification records available.

	PERFORMANCE REQUIREMENT FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.PR(01).2025
		Date : November 2025
SECTION C:	ASCERTAINED PERFORMANCE DEDUCTION (APD)	Section Revision: 2
		Page 13 of 21

ii. Execution Time Service Level as in Schedule A below:

No	Work Category	Description	Allocated Time
1	Emergency	• High risk of danger to safety & health of user/customer. Can cause imminent threat to life or property. • Situation which cause a disaster to core business process. • Loss of utilities at high rate. • The main objective is to control the situation and reduce damage/disturbance.	<1 hour (office hour) <2 hours (non-office hour)
2	Urgent	• Low risk of danger to safety & health of user/customer. • Situation which cause disturbance to core business process. • Can cause major operational disruption and problems if not attended quickly, or may develop to be Emergency category if not attended & remedied.	Within 2 days
3	Normal	• Situation which cause minimal impact to core business process. • Not immediately detrimental and not causing operational problem.	Within 3 days
4	Others	Lift Entrapment	Release within 15 minutes

Schedule A - Execution Time Service Level

iii. Percentage for numbers of Work Order that are not comply with the allocated Execution Time divided by total Work Order for the Evaluation Month shall be used to evaluate the PI achievement monthly. APD shall be imposed if the PI condition is not met.

3.2.1E Pending/Backlog Work Order

- Pending/Backlog Work Order can be traced in CMMS records, manual Work Order forms/sheet or other records available.
- Pending/Backlog Work Order as in Schedule B below:

	PERFORMANCE REQUIREMENT FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.PR(01).2025
		Date : November 2025
SECTION C:	ASCERTAINED PERFORMANCE DEDUCTION (APD)	Section Revision: 2
		Page 14 of 21

No	Pending Time	Target
1	Less than 30days	90% of Work Order
2	Less than 60days	95% of Work Order
3	Less than 90days	100% of Work Order

Schedule B–Pending/Backlog Work Order Service Level

- iii. Percentage of completed Pending/Backlog Work Orders divided by the total Pending/Backlog Work Orders shall be used to evaluate the PI achievement monthly. APD shall be imposed if the PI condition is not met.
- iv. FSO shall decide for further action regarding Pending/Backlog Work Orders that are still not completed (such as through appointing a third party to complete the work order) depending on the criticality and urgency by appropriate process permissible to the FSO.

3.2.1F Routine Inspection/Self-Finding

- i. Self-Finding Work Order record can be traced in CMMS records, Manual Work Order forms/sheet or other agreed completion verification records available.
- ii. Self-Finding Work Order quantity shall be more than 80% of the total Corrective Work Order of the Evaluation Month. APD shall be imposed if the PI condition is not met.

Custodial Services Performance

- a) Custodial Services in this contract will be done by the Contractor or Sub-Contractor appointed/nominated by FSO/Client through Provisional Sum item in the Contract or directly appointed by FSO/Client.
- b) If the Custodial Services done by the Contractor, the responsibility of achieving the PI shall be liable to the Contractor and the APD shall be imposed to the Contractor if the PI condition is not met.
- c) Whereas, if the Custodial Services is done by the appointed Sub-Contractor and direct payments from the Government is opted (e.g. through Deed of Assignment method), in the event of not achieving the PI, the calculations of APD shall be based on the Monthly Payment Value (MPV) of the Custodial Services item only and shall be imposed to the Custodial Services Sub-Contractor directly. The Contractor's Profit & Attendance shall be deducted pro-rated according to APD value of the Custodial Services.

	PERFORMANCE REQUIREMENT FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.PR(01).2025
		Date : November 2025
SECTION C:	ASCERTAINED PERFORMANCE DEDUCTION (APD)	Section Revision: 2
		Page 15 of 21

3.2.1G Cleaning Performance

- i. Walkthrough audit by FSO/FSOR team to make sampling on Housekeeping/Cleaning Work Order in the Evaluation Month against the site condition.
- ii. The performance level of cleaning shall follow MS2550:2014 (or latest publication) evaluation method. The Contractor is responsible to get a copy of MS2550 for the usage throughout the Contract period.
- iii. The average marks (percentage) from total numbers of walkthrough audit evaluation marks shall be used to evaluate the PI achievement monthly. APD shall be imposed to the Contractor if the PI condition is not met.

3.2.1H Pest Control Performance

- i. Walkthrough audit by FSO/FSOR team to make sampling on Pest Control Work Order in the Evaluation Month against the site condition.
- ii. CMMS or other records regarding the quality of pest control services shall also be used as the performance evaluation.
- iii. The completed Pest Control Work Order divided by the total Pest Control Work Order from walkthrough audit of the Evaluation Month shall be used to evaluate the PI achievement monthly. APD shall be imposed to the Contractor if the PI condition is not met.

Service Availability

3.2.1I Critical Services

- i. Critical services that will severely affect/interrupt the core business process such as electrical supply, air conditioning, water supply and lift services.
- ii. CMMS or other records regarding the downtime of the critical service shall be considered as non-availability of critical services.
- iii. Numbers of Daftar Aset Khusus (DAK) Ruang that are affected by downtime for the Evaluation Month divided by total DAK Ruang of the premise shall be used to evaluate the PI monthly. APD shall be imposed to the Contractor if the PI condition is not met.

	PERFORMANCE REQUIREMENT FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.PR(01).2025
		Date : November 2025
SECTION C:	ASCERTAINED PERFORMANCE DEDUCTION (APD)	Section Revision: 2
		Page 16 of 21

3.2.1J Normal Services

- i. Normal services that affect/interrupt support process such as toilet and lighting.
- ii. CMMS or other records regarding the downtime of normal service shall be considered as non-availability of normal services.
- iii. Numbers of Daftar Aset Khusus (DAK) Ruang that are affected by downtime for the Evaluation Month divided by total DAK Ruang of the premise shall be used to evaluate the PI monthly. APD shall be imposed to the Contractor if the PI condition is not met.

3.2.2 **Asset Performance**

3.2.2A NCR for Architecture, C&S and Landscape assets 100% assets

- i. Non-Compliance Record (NCR) issued by FSO/FSOR team through random condition inspection/walkthrough audit. This inspection/audit shall make sampling on Architecture, C&S and Landscape PPM records in CMMS/manual records. NCR is issued when the actual site condition is not conforming to records.
- ii. NCR shall be issued through CMMS or other agreed means of notifications e.g. emails, written memo/letters, etc. to the Contractor. The Contractor shall response and rectifies the NCR within 24 hours of receiving the NCR notification.
- iii. The non-completed/overdue NCR shall be considered as non-performance to the PI. Numbers of non-completed/overdue NCR divided by total numbers of NCR for the Evaluation Month shall be used to evaluate the PI monthly. APD shall be imposed to the Contractor if the PI condition is not met.

3.2.2B NCR for Mechanical assets

- i. Non-Compliance Record (NCR) issued by FSO/FSOR team through random condition inspection/walkthrough audit. This inspection/audit shall make sampling on Mechanical PPM records in CMMS/manual records. NCR is issued when the actual site condition is not conforming to records.
- ii. NCR shall be issued through CMMS or other agreed means of notifications e.g. emails, written memo/letters, etc. to the Contractor. The Contractor shall response and rectifies the NCR within 24 hours of receiving the NCR notification.
- iii. The non-completed/overdue NCR shall be considered as non-performance to the PI. Numbers of non-completed/overdue NCR divided by total numbers of NCR for

	PERFORMANCE REQUIREMENT FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.PR(01).2025
		Date : November 2025
SECTION C:	ASCERTAINED PERFORMANCE DEDUCTION (APD)	Section Revision: 2
		Page 17 of 21

the Evaluation Month shall be used to evaluate the PI monthly. APD shall be imposed to the Contractor if the PI condition is not met.

3.2.2C NCR for Electrical assets

- i. Non-Compliance Record (NCR) issued by FSO/FSOR team through random condition inspection/walkthrough audit. This inspection/audit shall make sampling on Electrical PPM records in CMMS/manual records. NCR is issued when the actual site condition is not conforming to records.
- ii. NCR shall be issued through CMMS or other agreed means of notifications e.g. emails, written memo/letters, etc. to the Contractor. The Contractor shall response and rectifies the NCR within 24 hours of receiving the NCR notification.
- iii. The non-completed/overdue NCR shall be considered as non-performance to the PI. Numbers of non-completed/overdue NCR divided by total numbers of NCR for the Evaluation Month shall be used to evaluate the PI monthly. APD shall be imposed to the Contractor if the PI condition is not met.

3.2.2D Engineering Reports & Recommendation Action

- i. Engineering reports that need further action taken for example, Competent Electrical Engineer report recommend renewing the registration of Electrical Installation, but the Contractor fails to prove the action (i.e. Work Order, work done report, picture) to the FSO/FSOR, the report shall be considered as a non-performance to the PI.

3.2.2E NCR for Work-Done as Specification / Asset Quality Meet Standards

- i. Non-Compliance Record (NCR) issued by FSO/FSOR team through random condition inspection/walkthrough audit on the asset.
- ii. NCR shall be issued through CMMS or other agreed means of notifications e.g. emails, written memo/letters, etc. to the Contractor. The Contractor shall response and rectifies the NCR within 24 hours of receiving the NCR notification.
- iii. The non-completed/overdue NCR shall be considered as non-performance to the PI. Numbers of non-completed/overdue NCR divided by total numbers of NCR for the Evaluation Month shall be used to evaluate the PI monthly. APD shall be imposed to the Contractor if the PI condition is not met.

	PERFORMANCE REQUIREMENT FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.PR(01).2025
		Date : November 2025
SECTION C:	ASCERTAINED PERFORMANCE DEDUCTION (APD)	Section Revision: 2
		Page 18 of 21

3.2.3 Building Efficiency

3.2.3A Energy Saving Measure (ESM) Programs

- i. The ESM programs proposed in the Energy Audit Report and any WSM initiatives identified through water consumption analysis or usage trend assessment can be used as the ESM/WSM program list for FSO approval to be implemented throughout the Contract period.
- ii. The ESM/WSM program list can be amended accordingly subject to FSO approval.
- iii. The ESM/WSM program implementation shall be measured monthly. Any ESM/WSM program which cannot be implemented subject to valid justification, the Contractor shall write officially to get the approval from FSO/FSOR before the cut-off date.
- iv. The ESM/WSM program which is not being implemented for the Evaluation Month (considered non-performance to the PI) divided by the total numbers of planned ESM/WSM program for the Evaluation Month shall be used to evaluate the PI monthly. APD shall be imposed to the Contractor if the PI condition is not met.

3.2.3B Building Energy Intensity (BEI)

- i. Baseline BEI for the site shall be established after the energy audit. If an energy audit is not included or priced in the contract, the Building Energy Intensity (BEI) baseline shall be established using any bills related to energy consumption for 12 consecutive months, in accordance with the relevant standards and guidelines.
- ii. Proposed target BEI monthly shall be established towards achieving the proposed Yearly Energy Reduction Target. If the target cannot be achieved due to acceptable justification, the Contractor shall write officially to get FSO/FSOR's approval before the cut-off date.
- iii. The achievement of targeted monthly BEI shall be used to evaluate the PI monthly. APD shall be imposed to the Contractor if the PI condition is not met.

3.2.3C Utility Wastage

- i. Utility wastage for Electrical system (e.g. lighting, air conditioning), Cold Water supply (e.g. tank, water tap leak) and Chilled Water Supply (e.g. AHU, pumps running) that contributes to the unnecessary increase of utility bills shall be considered in this PI.
- ii. Utility Wastage is considered solely due to negligence of the Contractor in managing the operation & maintenance service.

	PERFORMANCE REQUIREMENT FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.PR(01).2025
		Date : November 2025
SECTION C:	ASCERTAINED PERFORMANCE DEDUCTION (APD)	Section Revision: 2
		Page 19 of 21

- iii. The CMMS records and walkthrough audit findings shall be the source of utility wastage records to evaluate the PI monthly. APD shall be imposed to the Contractor if the PI condition is not met.

3.2.4 Safety & Statutory Compliance

3.2.4A Comply Relevant Statutory Regulations

- i. The Contractor shall submit monthly reports regarding the status of the relevant statutory regulations as item 5.0 and 18.0 in Appendix 7A Technical Specification for Building Facility Management & Maintenance.
- ii. The Contractor shall write officially to get FSO/FSOR's considerations before the cut-off date if they fail to meet the relevant statutory regulations due to conditions outside of their control.
- iii. The compliance of relevant statutory regulations shall be used to evaluate the PI monthly. APD shall be imposed to the Contractor if the PI condition is not met.

3.2.4B HSE Programs

- i. The HSE programs proposal listed in the HSE Management Plan can be used as HSE program list for FSO approval to be used throughout the Contract period.
- ii. The HSE program list can be amended accordingly subject to FSO approval.
- iii. The HSE program implementation shall be measured monthly. Any HSE program which cannot be implemented subject to valid justification, the Contractor shall write officially to get the approval from FSO/FSOR before the cut-off date.
- iv. The HSE program which is not being implemented for the Evaluation Month (considered non-performance to the PI) divided by the total numbers of planned HSE program for the Evaluation Month shall be used to evaluate the PI monthly. APD shall be imposed to the Contractor if the PI condition is not met.

3.2.4C Reports submission

- i. The Contractor shall submit monthly reports as item 7.6 in Appendix 7A Technical Specification for Building Facility Management & Maintenance.
- ii. The reports shall be submitted on time before the cut-off date of the Evaluation Month, failing which shall be considered as a non-performance to the PI.
- iii. The content of the reports shall be evaluated to the sufficient requirements of FSO/FSOR. If the content is not up to the FSO/FSOR requirements, the reports

	PERFORMANCE REQUIREMENT FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.PR(01).2025
		Date : November 2025
SECTION C:	ASCERTAINED PERFORMANCE DEDUCTION (APD)	Section Revision: 2
		Page 20 of 21

shall be considered as a non-performance to the PI. APD shall be imposed to the Contractor if the PI condition is not met.

3.3 APD Calculation

- a) The Monthly Payment Value (MPV) shall be the basis for APD calculation.
- b) MPV is the actual work value that shall exclude Preliminaries, Management & Supervisory Staff Remuneration, Provisional Sum and Provisional Quantity items.
- c) The maximum APD (APD Max) value shall be **5%*** from MPV. The APD shall be calculated through the weightage for each PI.
- d) The weightage for each PI is multiplying by the APD Max value to get the value for each PI.
- e) The target achievement for each PI is a pass/fail condition whereby if the Contractor fails to meet the target specified for each, the APD value is being calculated.

*APD Max value can be adjusted accordingly

	PERFORMANCE REQUIREMENT FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.PR(01).2025
		Date : November 2025
SECTION C:	ASCERTAINED PERFORMANCE DEDUCTION (APD)	Section Revision: 2
		Page 21 of 21

3.4 APD Calculation Samples

APD Calculation

Evaluation Month
 Monthly Payment Value (MPV)
 Maximum APD (%)
 APD Max

Oct-25	(fill)
RM500,000.00	(fill)
5	(fill)
RM25,000.00	

No.	Key Performance Indicator (KPI)	PI No.	Performance Indicator (PI)	Target* (%)	Actual (%)	Demerit Point*	Demerit Points Imposed	Weightage* (%)	APD Value (RM)	APD Value Deducted (RM)
1	FMM Service Delivery related to Core Business	1A	Customer Satisfaction Survey (CSS)	80	75.00	6	6	5	RM1,250.00	RM1,250.00
		1B	Customer Rating in Work Order Sheet	70	75.00	6	0	5	RM1,250.00	RM0.00
		1C	Response Time	100	100.00	4	0	5	RM1,250.00	RM0.00
		1D	Execution Time	95	87.00	6	6	5	RM1,250.00	RM1,250.00
		1E	Pending/Backlog Work Order Completion	100	100.00	4	0	5	RM1,250.00	RM0.00
		1F	Self-finding Work Order≥80% from Total Work Order	80	58.00	4	4	10	RM2,500.00	RM2,500.00
		1G	Cleaning Performance	85	90.00	5	0	6	RM1,500.00	RM0.00
		1H	Pest Control Performance	95	97.00	5	0	5	RM1,250.00	RM0.00
		1I	Critical Services	95	99.67	7	0	8	RM2,000.00	RM0.00
		1J	Normal Services	85	85.00	4	0	7	RM1,750.00	RM0.00
2	Asset Performance	2A	PPM for Architecture, C&S and Landscape assets implemented	100	100.00	8	0	4	RM1,000.00	RM0.00
		2B	PPM for Mechanical asset implemented	100	95.00	8	8	4	RM1,000.00	RM1,000.00
		2C	PPM for Electrical asset implemented	100	85.00	8	8	4	RM1,000.00	RM1,000.00
		2D	Engineering/System Reports & Recommendation action taken	100	99.00	3	3	4	RM1,000.00	RM1,000.00
		2E	Work done as specification and asset quality meet standards	100	79.00	3	3	4	RM1,000.00	RM1,000.00
3	Building Energy Efficiency	3A	Energy/water Conservation programs implemented	100	100.00	4	0	4	RM1,000.00	RM0.00
		3B	Building Energy Intensity (BEI) target met	100	100.00	3	0	3	RM750.00	RM0.00
		3C	Utility Consumption No wastage	100	94.00	3	3	3	RM750.00	RM750.00
4	Safety & Statutory Compliance	4A	Relevant Acts & Regulations Compliance	100	100.00	3	0	3	RM750.00	RM0.00
		4B	HSE programs implemented	100	100.00	3	0	3	RM750.00	RM0.00
		4C	Reports submitted on time with sufficient content	100	100.00	3	0	3	RM750.00	RM0.00
						100	41	100	RM25,000.00	RM9,750.00

	PERFORMANCE REQUIREMENT FOR BUILDING FACILITY MANAGEMENT & MAINTENANCE	CPAB.BPPA.FMM.PR(01).2025
		Date : November 2025
SECTION D:	DEMERIT POINTS	Section Revision: 2
		Page 22 of 21

SECTION D: DEMERIT POINTS

4.1 Demerit Points Imposition

- a) Demerit points shall be imposed when the Contractor fails to meet the Performance Indicator (PI). Each PI carries demerit point depending on severity of the failure. The demerit points are accumulated every month.
- b) Warning letter is a letter to reprimand a contractor when the services provided are unsatisfactory based on the contract's requirements.
- c) Warning/Reminder Letter could be issue by the FSO if the performance of the contractor did not meet the FSO requirement. As for example, Warning/Reminder Letter could be issue by the FSO if but not limited to Demerit point exceed 40% for the month.
Notice of Remedy could be issued if lack of action taken by the contractors when Warning/Reminder letter was issued, or the actions does not up to FSO satisfactory.